CS177 Bioinformatics: Software Development and Applications

Lecture 1: Basics of Molecular Biology

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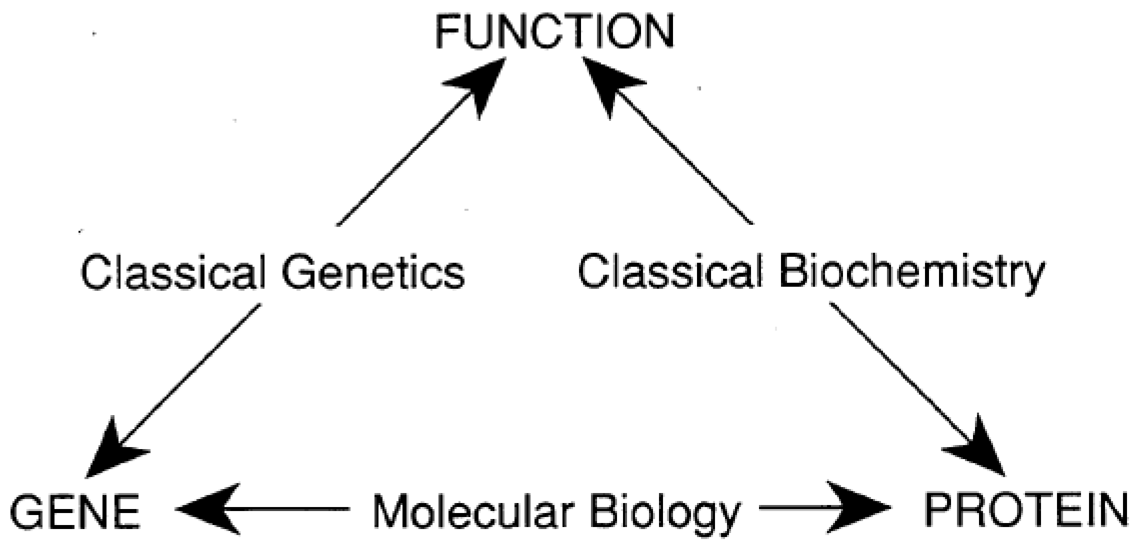
Learning objectives

- At the end of this lecture, you will be able to:
  - Explain basic terminology of molecular biology
  - Describe the basic building blocks of life
  - Describe the Central Dogma of Molecular Biology, i.e. how information is transmitted from DNA to proteins
  - Describe the processes of RNA transcription and protein translation

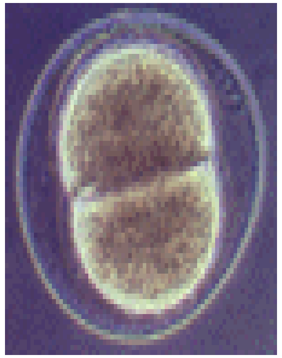
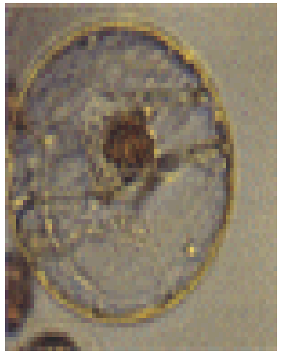
Historical roots of molecular biology

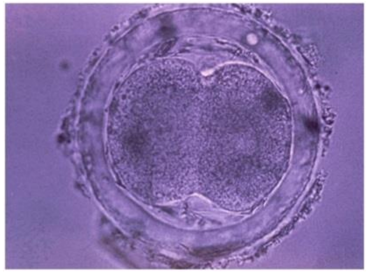
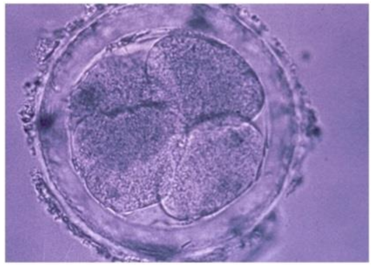
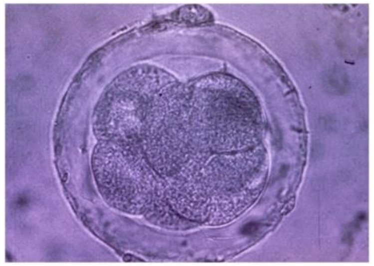


From Fig. 1.10, Lander and Waterman's chapter (1995)

- **Molecular biology** grew out of two experimental approaches to studying biological function: "Biochemistry" and "Genetics"
  - **Biochemistry**: break up the molecules in a living organism, with the goal of purifying and characterizing the chemical components responsible for carrying out a particular function
  - **Genetics**: study mutant organisms that are intact except for a single component

What Is Life Made of?

Life begins with Cell

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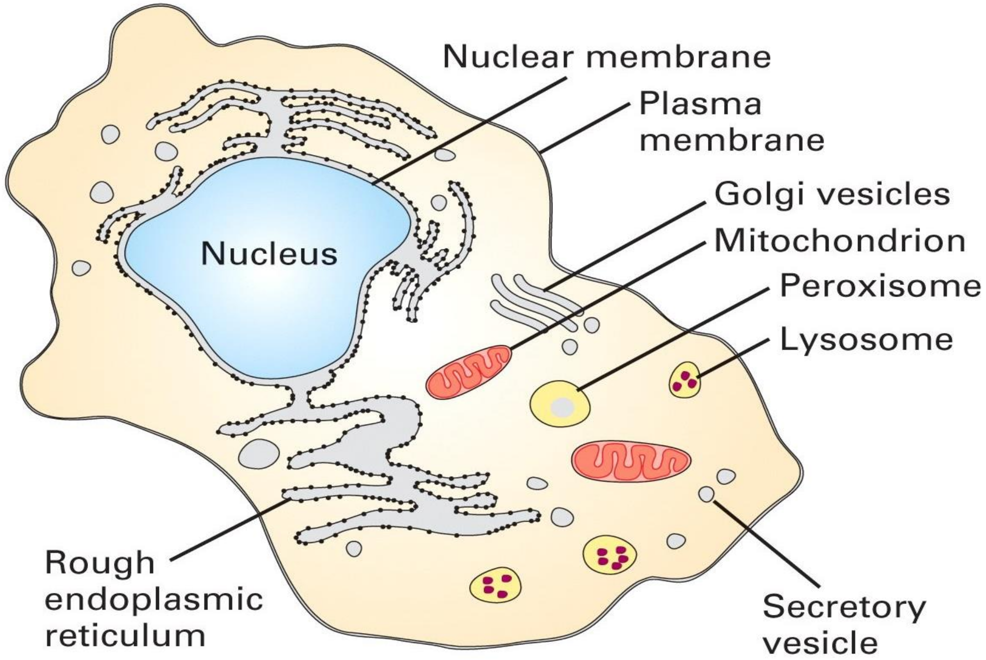

- A cell is a smallest structural unit of an organism that is capable of independent functioning
- All cells have some common features

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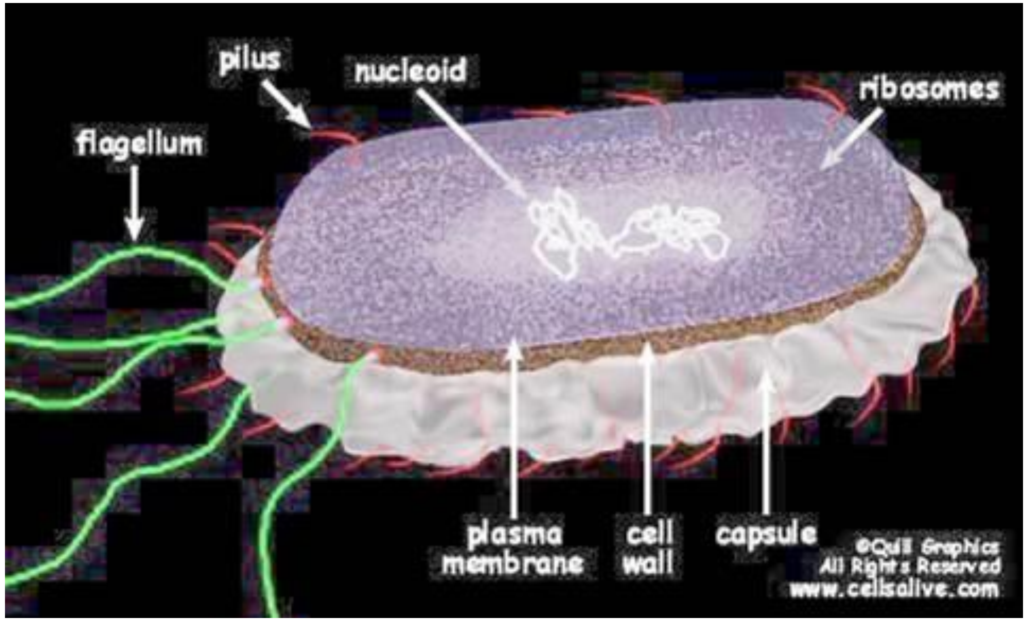
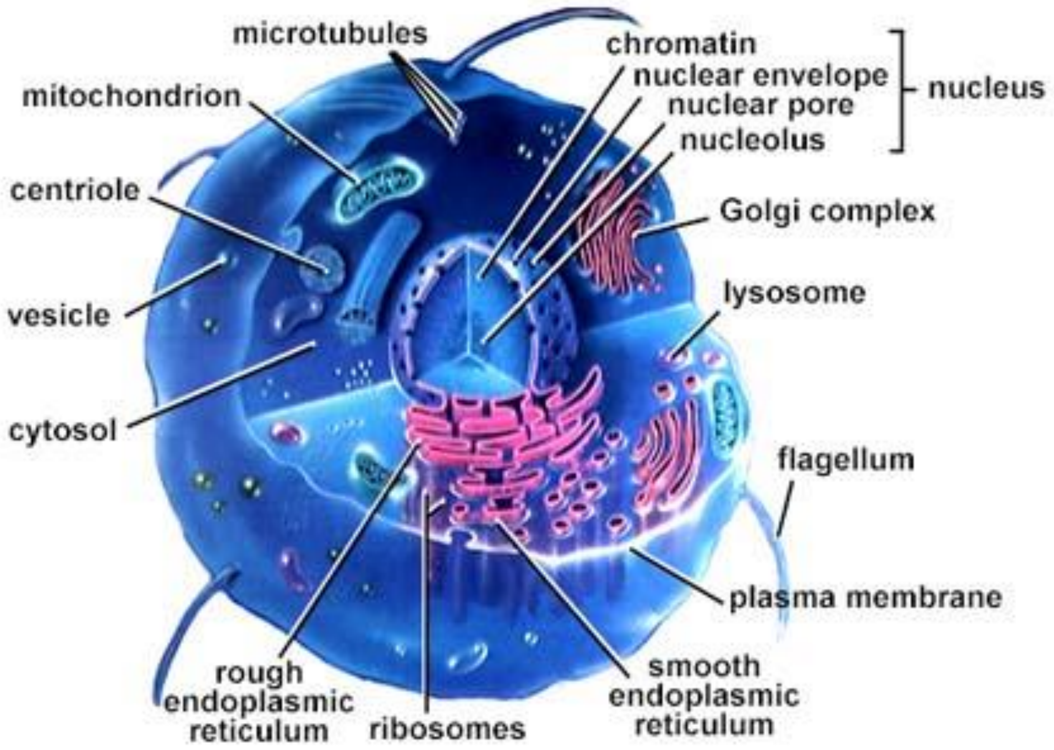
What is a cell made of?

- **Chemical composition** - by weight
  - **70%** water
  - **7%** small molecules
    - ➢ salts
    - ➢ lipids
    - ➢ amino acids
    - ➢ nucleotides
  - **23%** macromolecules
    - ➢ proteins
    - ➢ polysaccharides
    - ➢ lipids

Types of cells: Prokaryotes vs. Eukaryotes



A prokaryotic cell

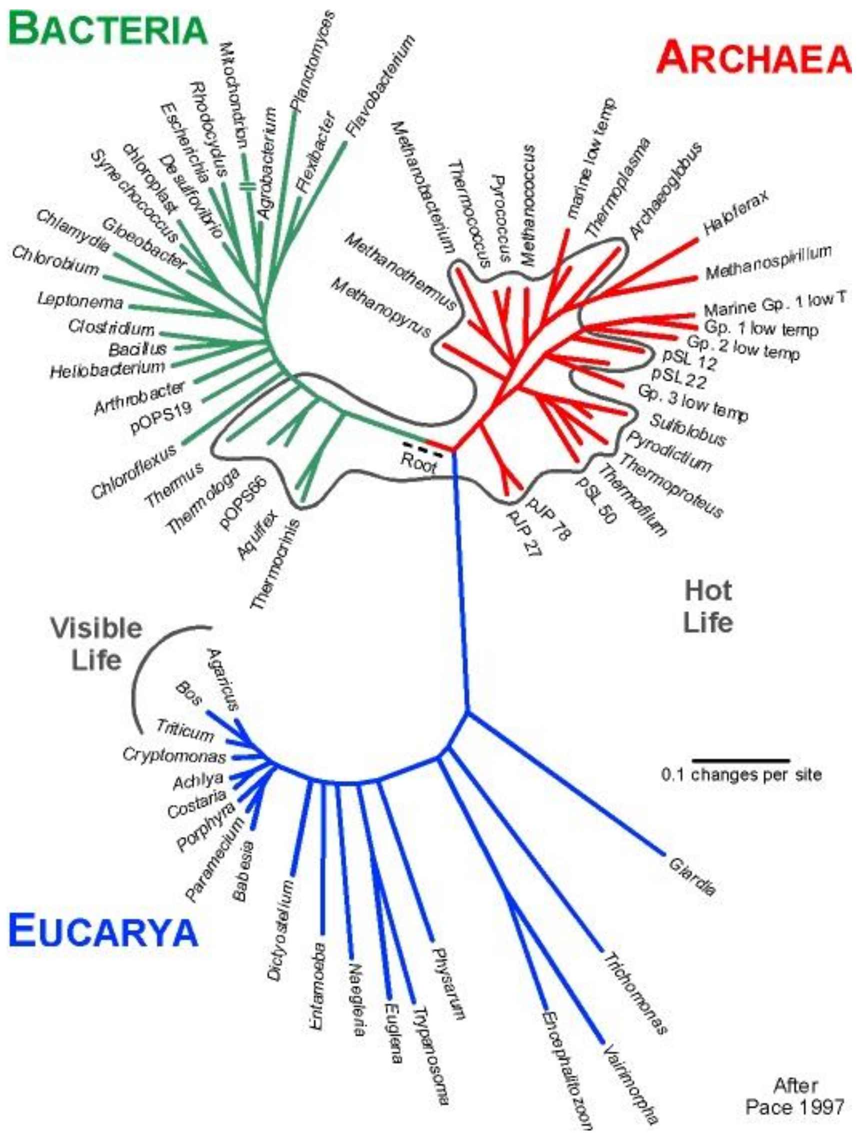


A eukaryotic cell

Prokaryotes vs. Eukaryotes

Prokaryotes	**Eukaryotes**
Single cell	Single or multi cell
No nucleus	Nucleus
No organelles	Organelles
One piece of circular DNA	Chromosomes
No mRNA post transcriptional modification	Exons/Introns splicing

Kingdoms of life



- According to the most recent evidence, there are three main branches in the tree of life.
- Prokaryotes include Archaea (“ancient ones”) and bacteria.
- Eukaryotes are kingdom Eukarya which includes plants, animals, fungi and certain algae.

From Pace NR (1997)
Science 276:734

Terminology

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- **Genome**: an organism's genetic material
  - The **genome** is an organism's complete set of DNA
    - A bacteria contains about 600,000 DNA base pairs (bp)
    - Human and mouse genomes have some 3 billion bp
  - The human genome has 24 *distinct* **chromosomes**
    - 22 autosomal chromosome pairs, plus sex-determining X and Y chromosomes
    - Each chromosome contains many **genes**
- **Gene**: a discrete unit of hereditary information located on the chromosomes and consisting of DNA
  - Basic physical and functional units of heredity
  - Specific sequences of DNA bases that encode instructions on how to make **proteins**
- **Nucleotides** are biological molecules that form the building blocks of nucleic acids (DNA and RNA)

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Terminology (continue)

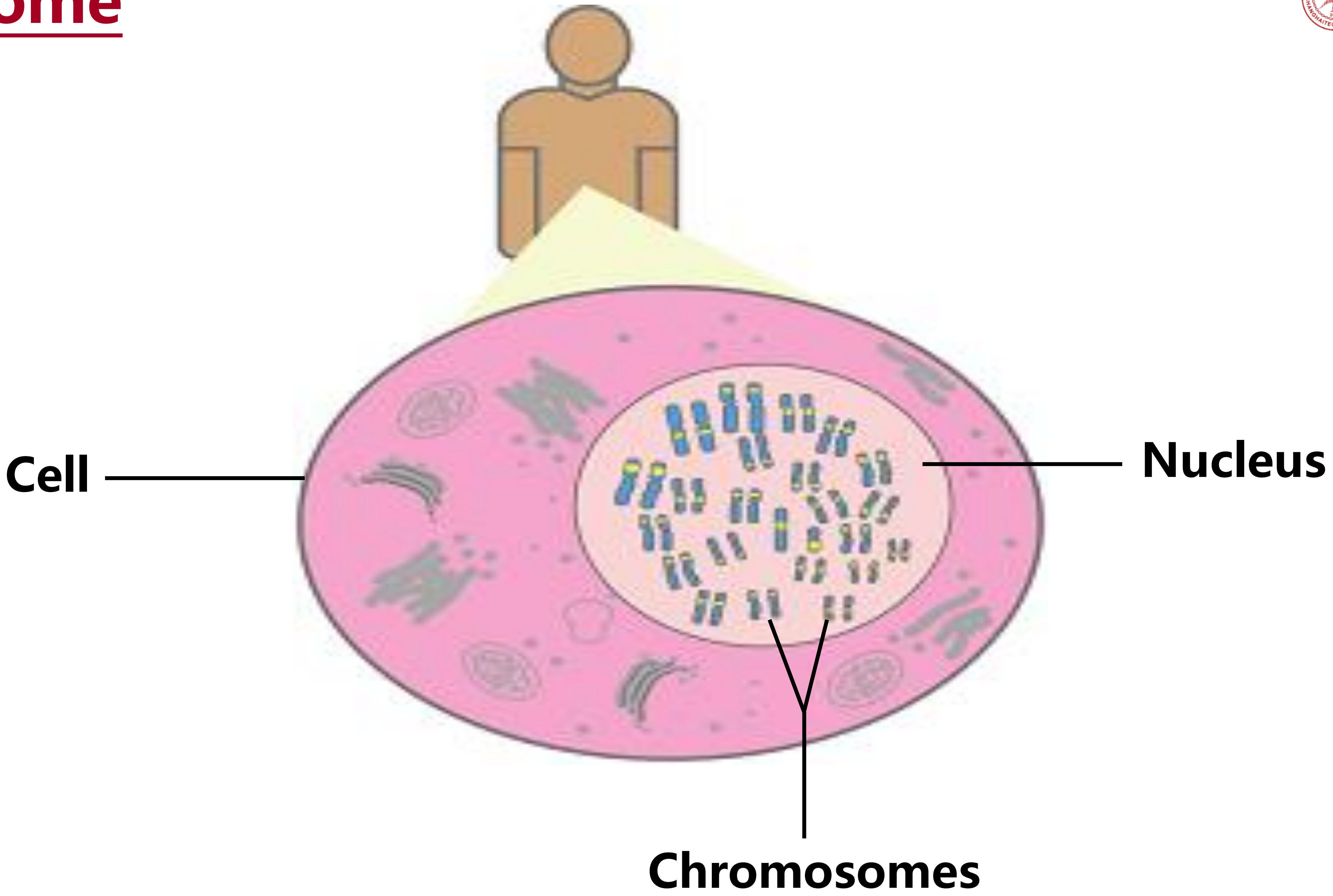
- **Proteins**
  - Make up the cellular structure
  - Large, complex molecules made up of smaller subunits called **amino acids**

- **Amino acids** are organic compounds that combine to form proteins

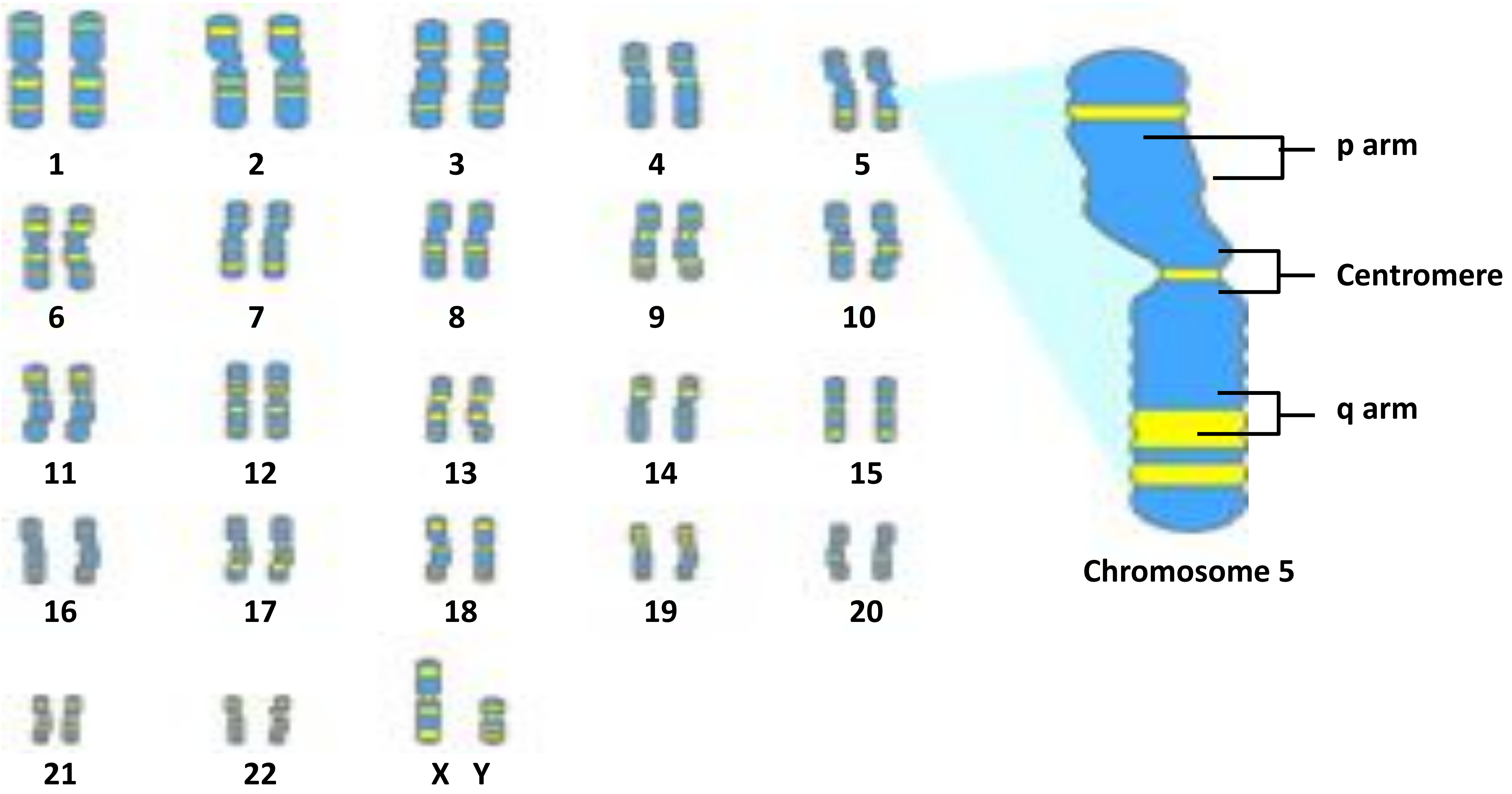
- **Genotype**: The genetic makeup of an organism

- **Phenotype**: the physical expressed traits of an organism

Genome



Chromosome



Chromosome 5

Overview of organizations of life

- **Nucleus = library**
- **Chromosomes = bookshelves**
- **Genes = books**
- Almost every cell in an organism contains the same libraries and the same sets of books.
- Books represent almost all the information (DNA) that every cell in the body needs so it can grow and carry out its various functions.

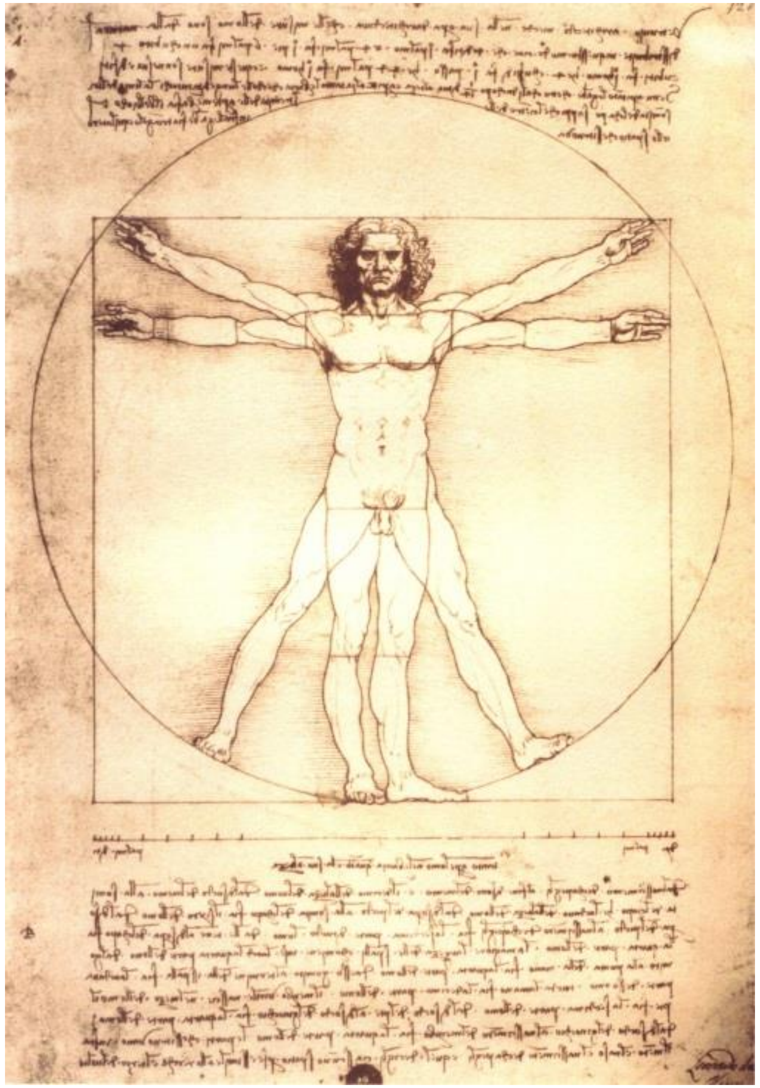
Cells information and machinery

- Cells store all information to replicate itself
  - Human genome is around 3 billion base pairs (bp) long
  - Almost every cell in human body contains the same set of genes
  - But not all genes are used or expressed by those cells

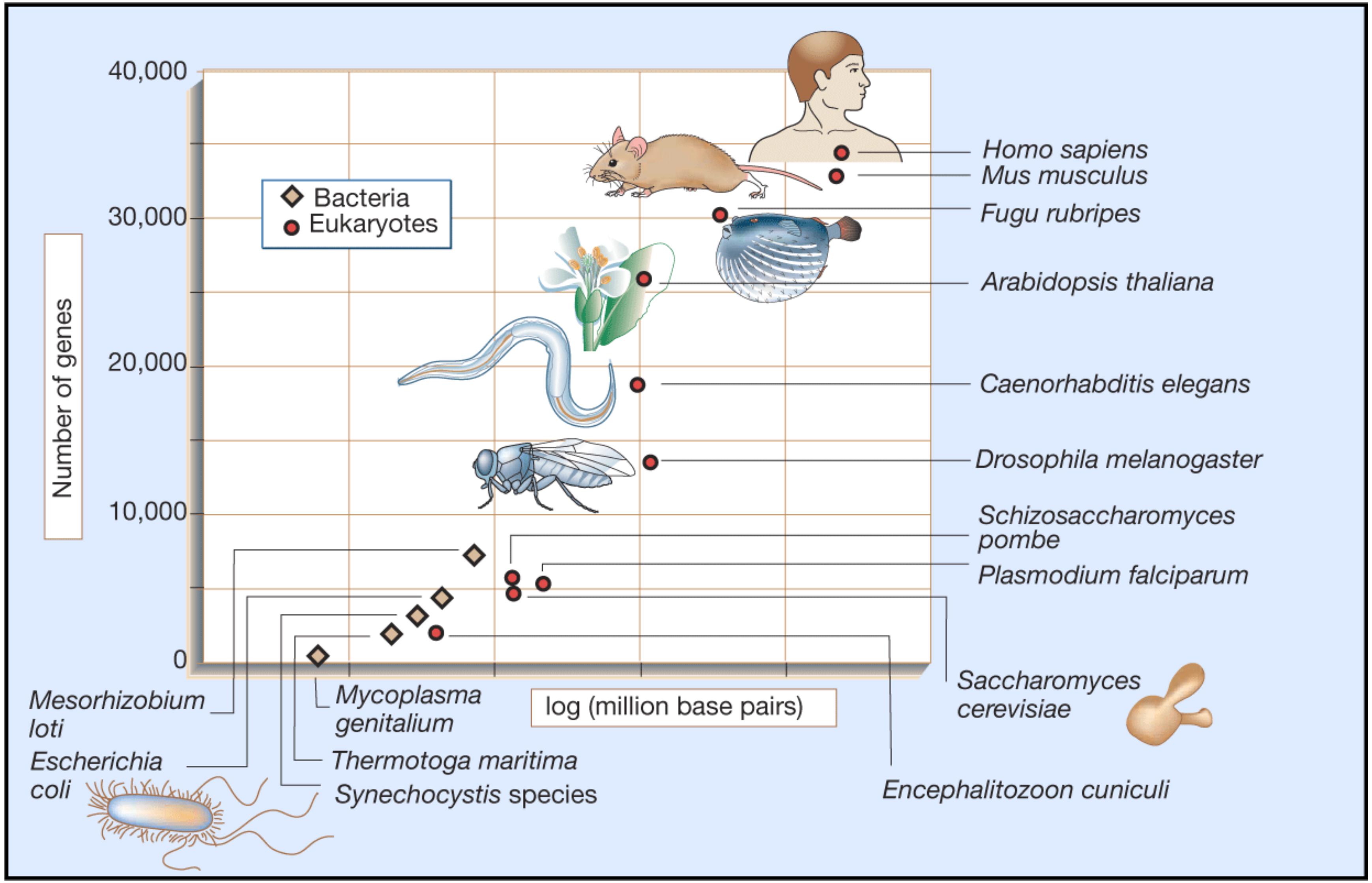
- Machinery:
  - Collect and manufacture components
  - Carry out replication
  - Kick-start its new offspring

Cells in the human body

- About 10^{12} cells in total
- Each cell
  - 23 pairs of chromosomes
  - 3.2 billion pairs of DNA bases
  - ~ 25,000 genes



Sizes of genomes and numbers of genes



The Central Dogma of Molecular Biology

All life depends on 3 critical molecules

- Deoxyribonucleic acid (**DNA**)
  - Hold information on how cell works
- Ribonucleic acid (**RNA**)
  - Act to transfer short pieces of information to different parts of cell
  - Provide templates to synthesize into protein
- **Protein**
  - Form enzymes that send signals to other cells and regulate gene activity
  - Form body’ s major components (e.g. hair, skin, etc.)

Watson & Crick – *"…the secret of life"*

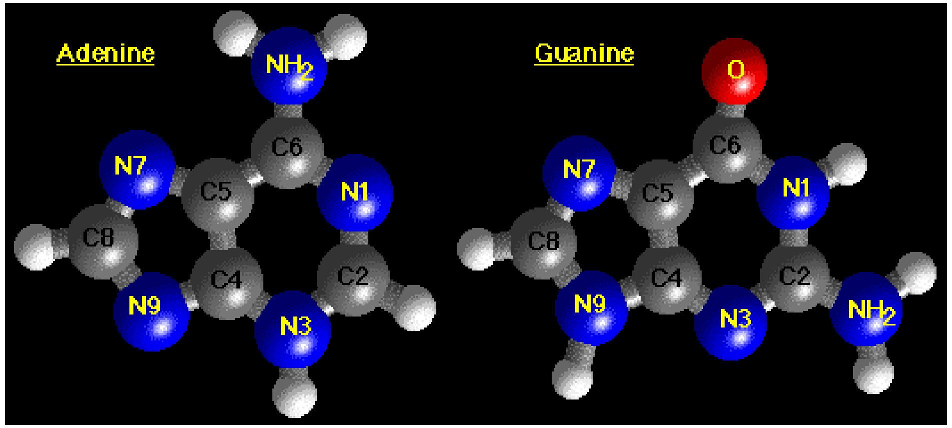
- J. Watson: a zoologist, F. Crick: a physicist
- "In 1947 Crick knew no biology and practically no organic chemistry or crystallography."
- Applying Chagraff's rules and the X-ray image from Rosalind Franklin, they constructed a "tinkertoy" model showing the double helix
- Their 1953 *Nature* paper: "It has not escaped our notice that the specific pairing we have postulated immediately suggests a possible copying mechanism for the genetic material."

Watson & Crick with DNA model

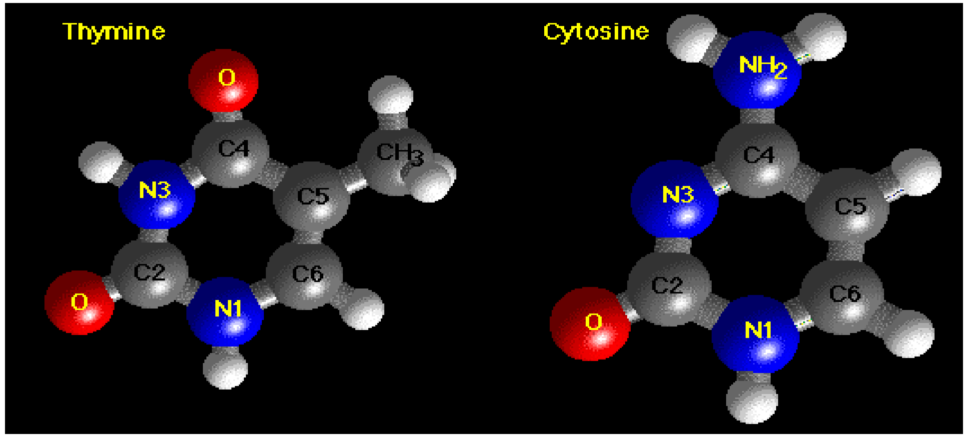
Rosalind Franklin with X-ray image of DNA

DNA base structure

- Structures of A & G (Purines)

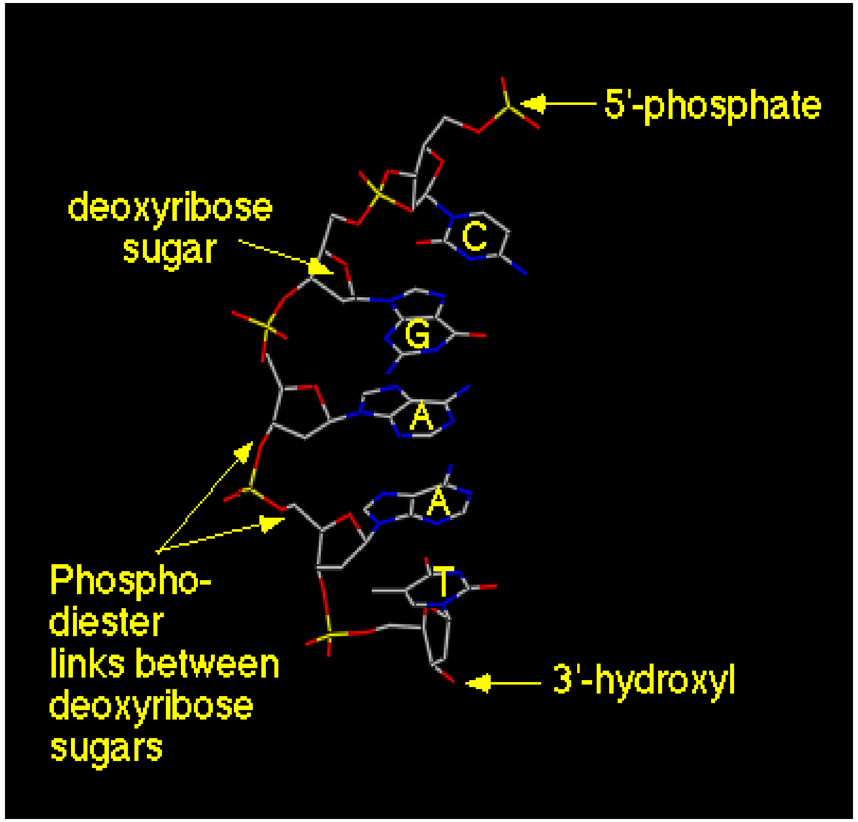


- Structures of T & C (Pyrimidines)



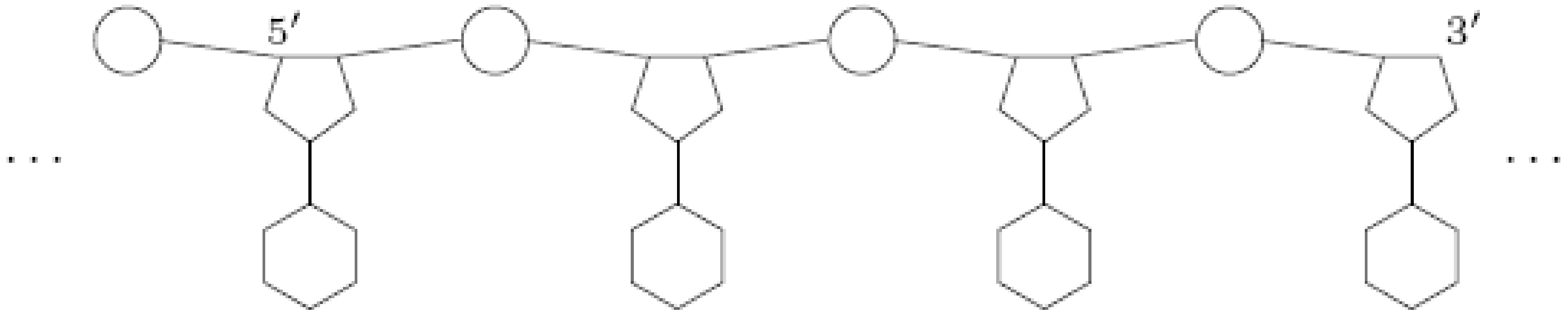
DNA backbone

- Alternating backbone of deoxyribose and phosphodiester groups
- Chain has a direction (known as **polarity**), 5'- to 3'- from top to bottom
- Oxygens (red atoms) of phosphates are polar and negatively charged
- A, G, C, and T bases extend away from chain, and stack on-top each other
- Bases are **hydrophobic**



Nucleic acid

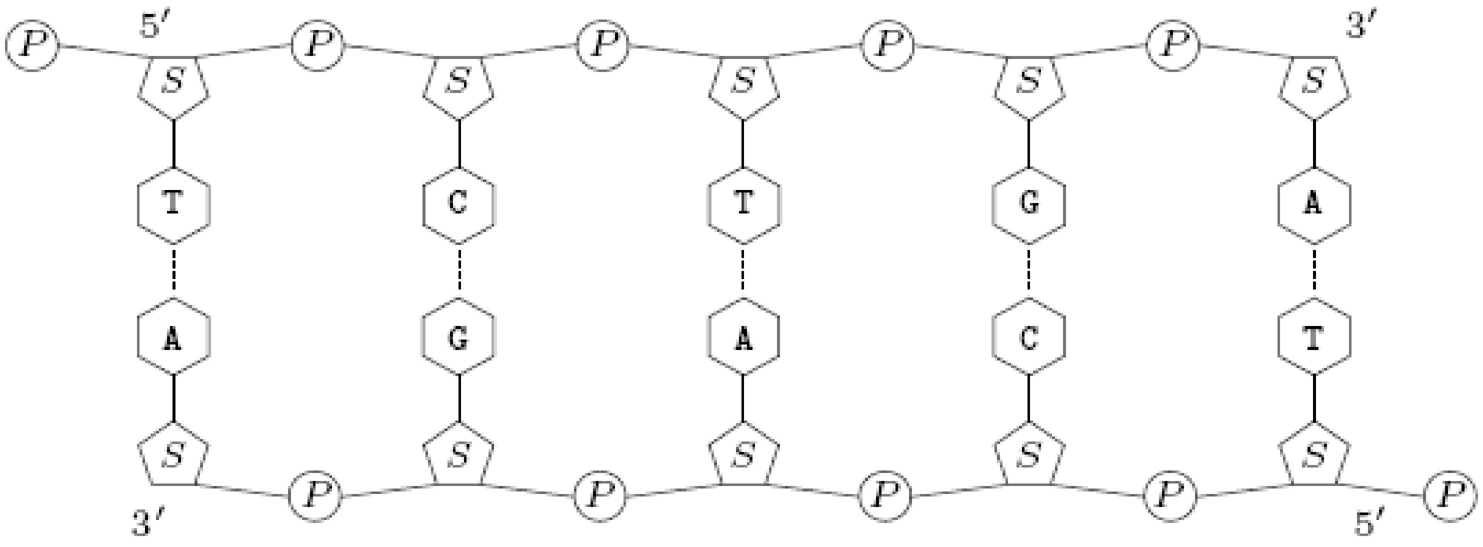
- Nucleic acids are responsible for encoding and storing the heredity information
- Schematic view of a nucleotide chain in the 5′ - 3′ direction
- Two different types of nucleic acids
  - Deoxyribonucleic acid (DNA)
  - Ribonucleic acid (RNA)



Example

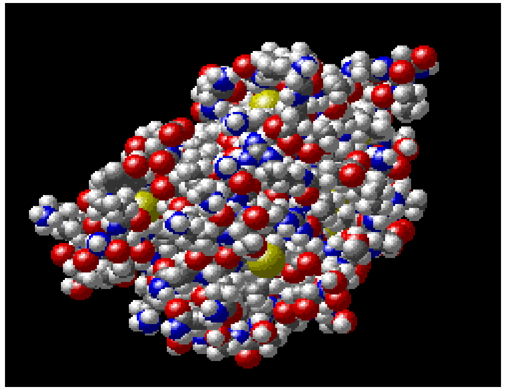
- Let s = TCTGA be the string representation of a DNA strand
- Then s~ = TCAGA is the reverse complement of s,
  - Unit of measurement: bp (standing for base pair)
  - By convention, DNA sequences are written in a 5′ -to-3′ direction

```
s    : 5′ : : : TCTGA : : : 3′
s~   : 3′ : : : AGACT : : : 5′
```



Genetic information

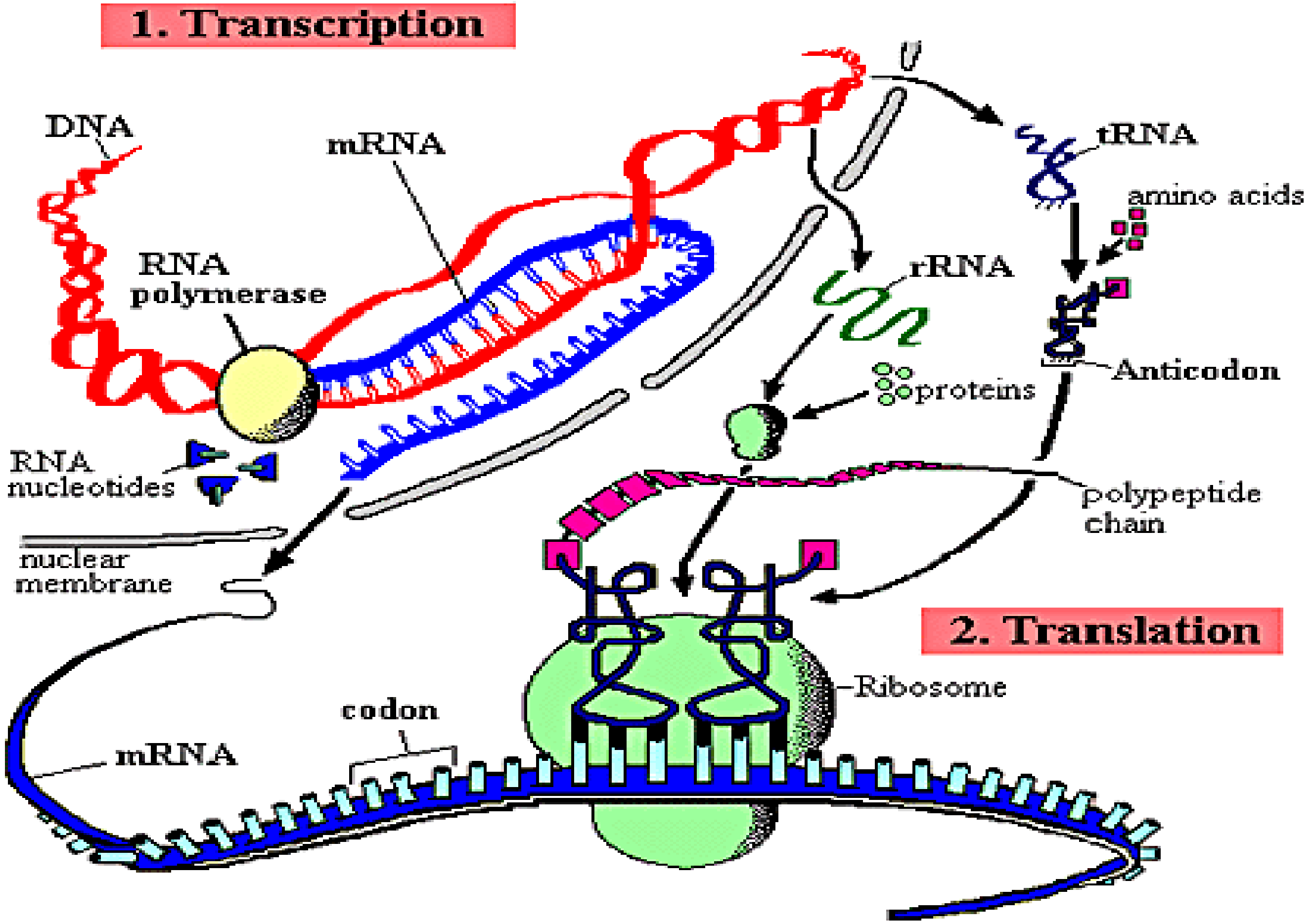
DNA $\xrightarrow{\text{transcription}}$ RNA $\xrightarrow{\text{translation}}$ Protein $\longrightarrow$ Phenotype



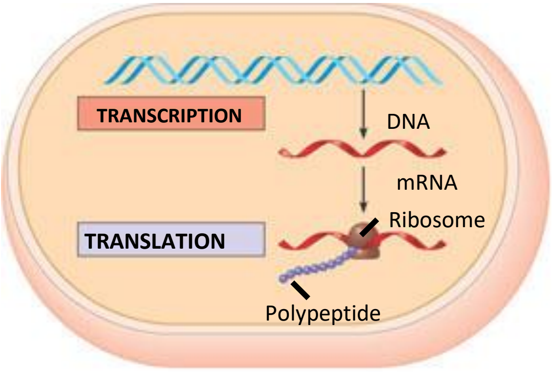
The Central Dogma of molecular biology

- **A fundamental principle of molecular biology**

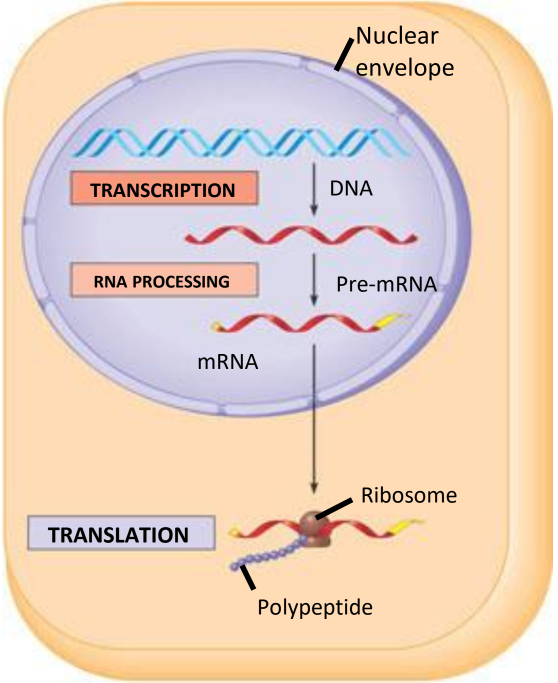
- Genetic information flows
  from DNA
  to RNA
  to protein



Transcription and translation in the flow of genetic information

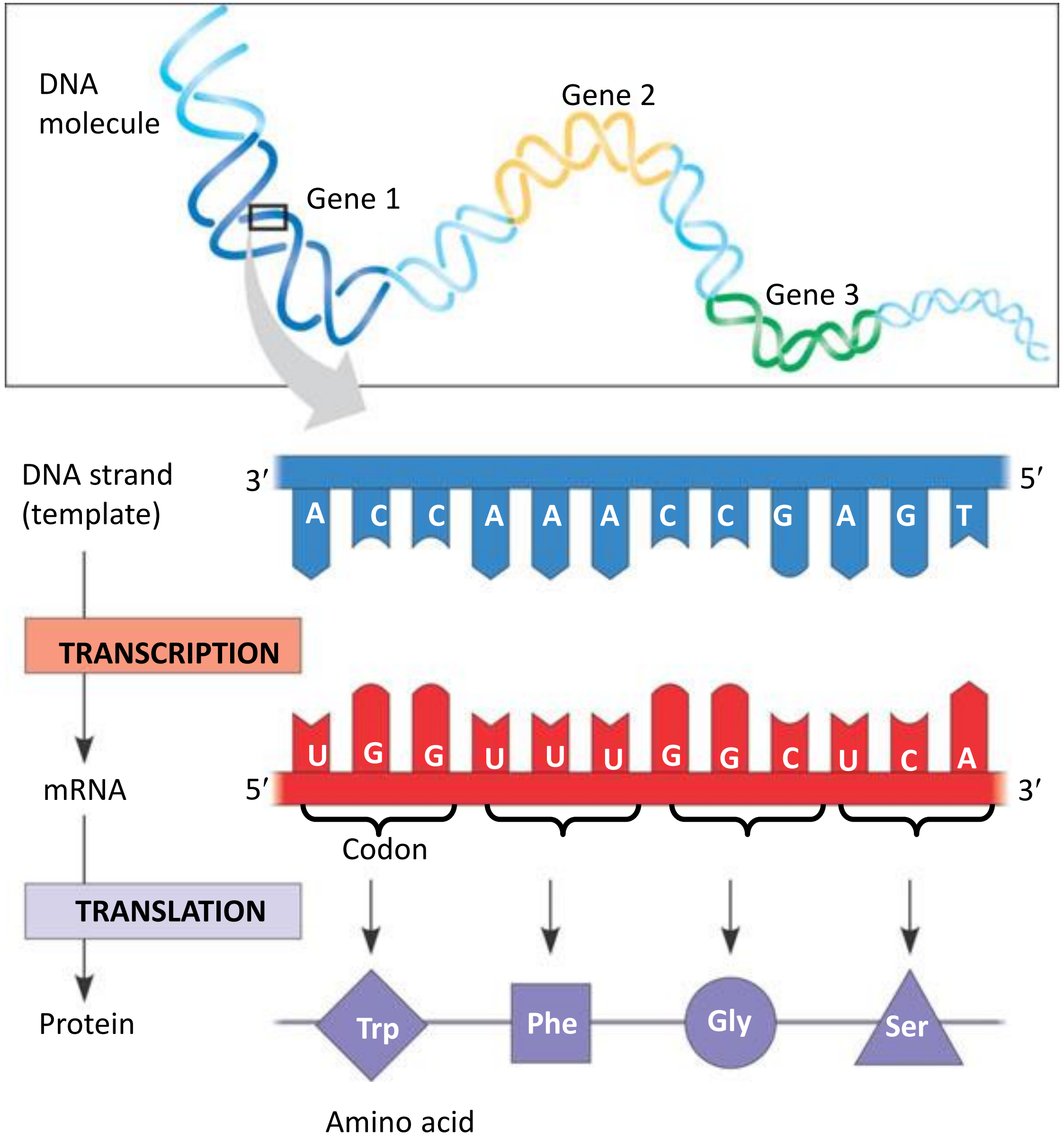


(a) **Prokaryotic cell.** In a cell lacking a nucleus, mRNA produced by transcription is immediately translated without additional processing.



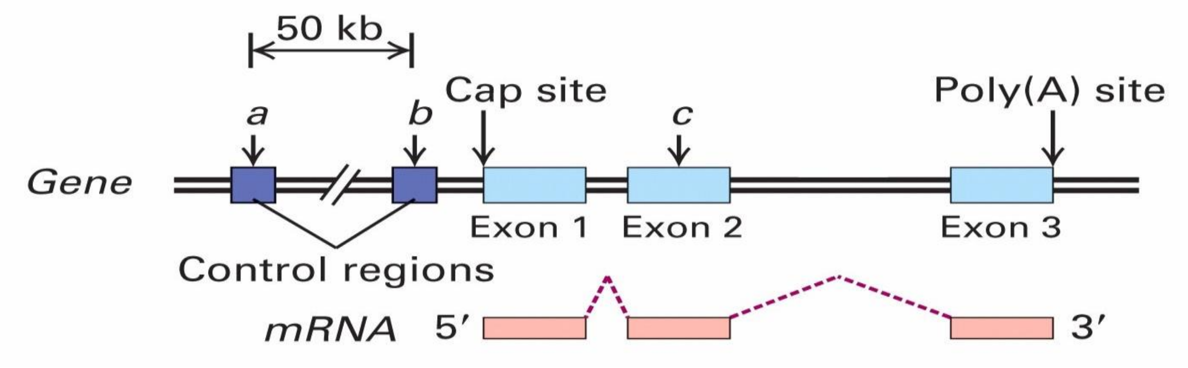
(b) **Eukaryotic cell.** The nucleus provides a separate compartment for transcription. The original RNA transcript, called pre-mRNA, is processed in various ways before leaving the nucleus as mRNA.

The triplet code



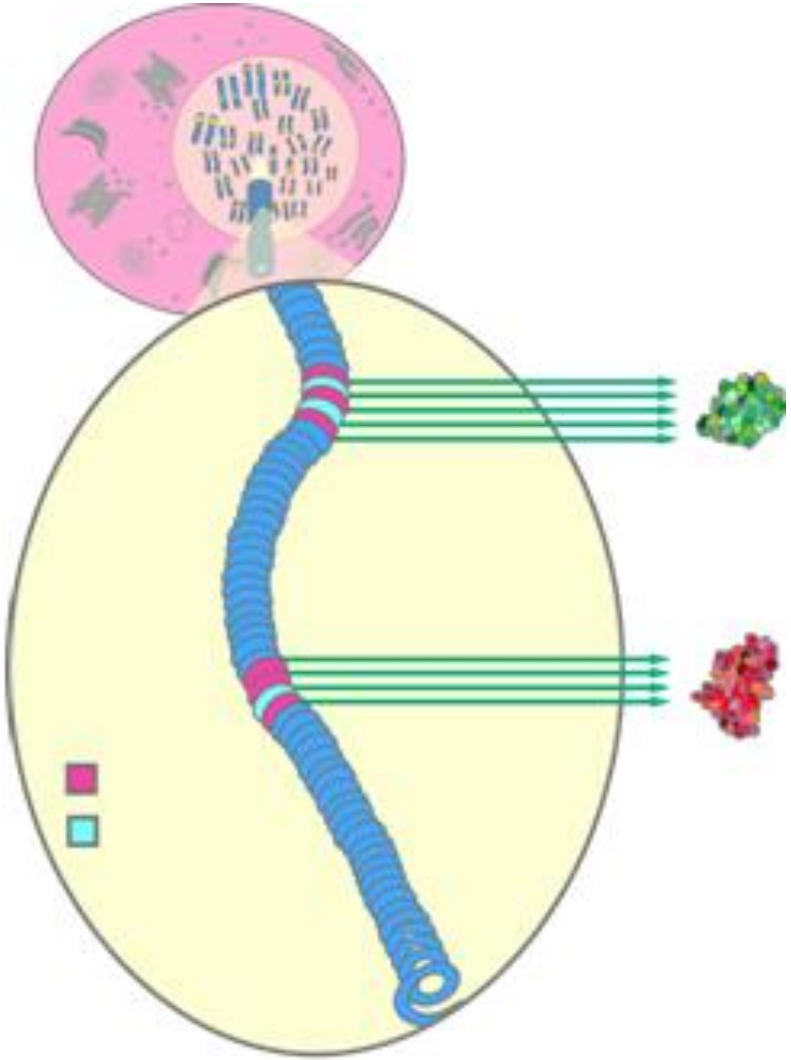
Transcriptional Regulation

Structure of a gene



- Regulatory regions: up to 50 kb upstream of +1 site
- Exons: Protein coding and untranslated regions (UTR)
  1 to 178 exons per gene (mean 8.8)
  8 bp to 17 kb per exon (mean 145 bp)
- Introns: Splice acceptor and donor sites, junk DNA
  Average 1 kb – 50 kb per intron
- Gene size: Largest – 2.4 Mb (Dystrophin). Mean – 27 kb.

Noncoding regions



Gene 1

Gene 2

Protein 1

Noncoding region

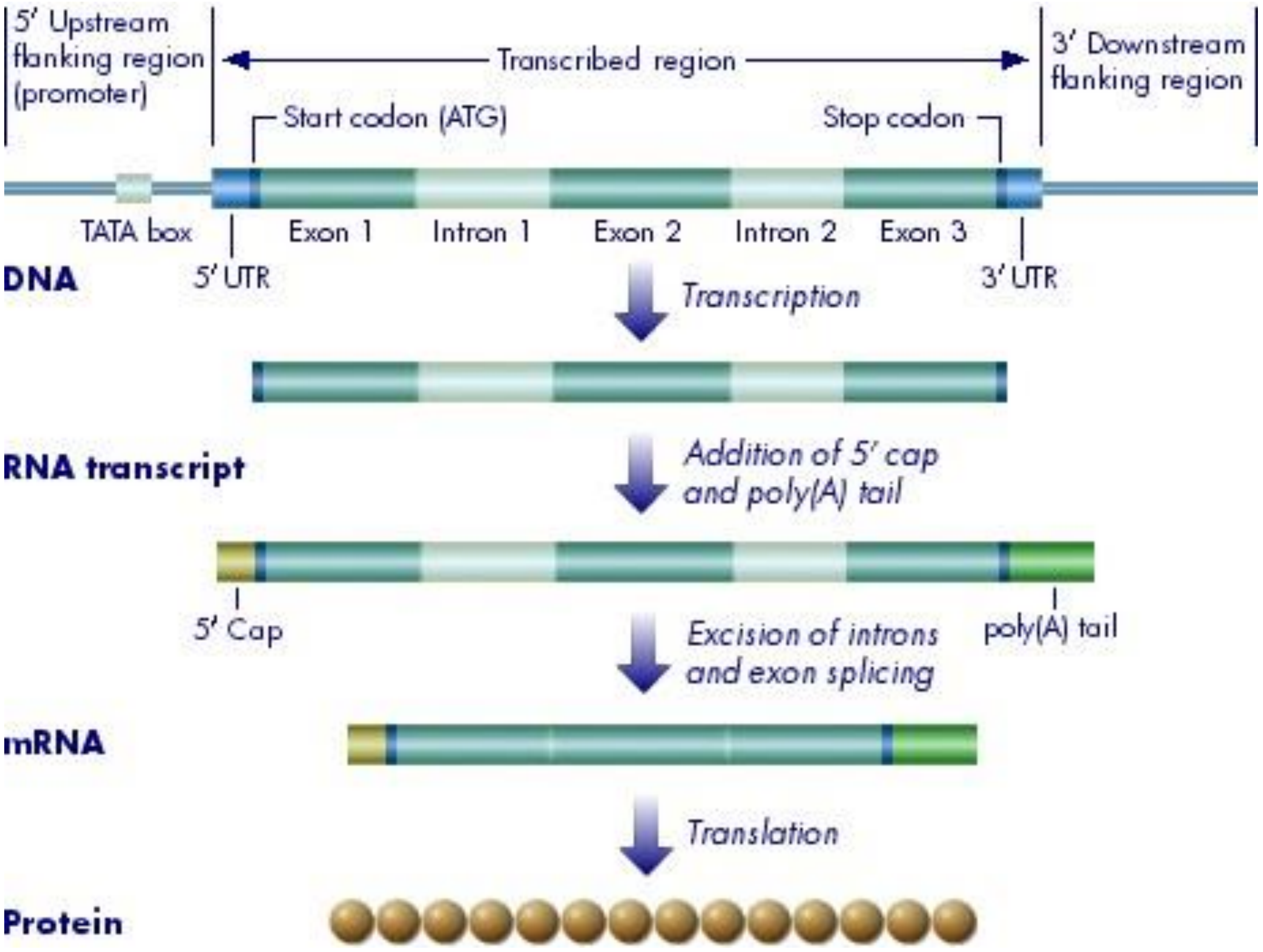
Protein 2

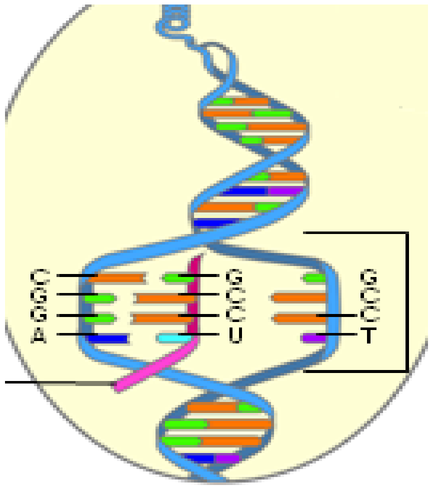
Noncoding region

mRNA transcription

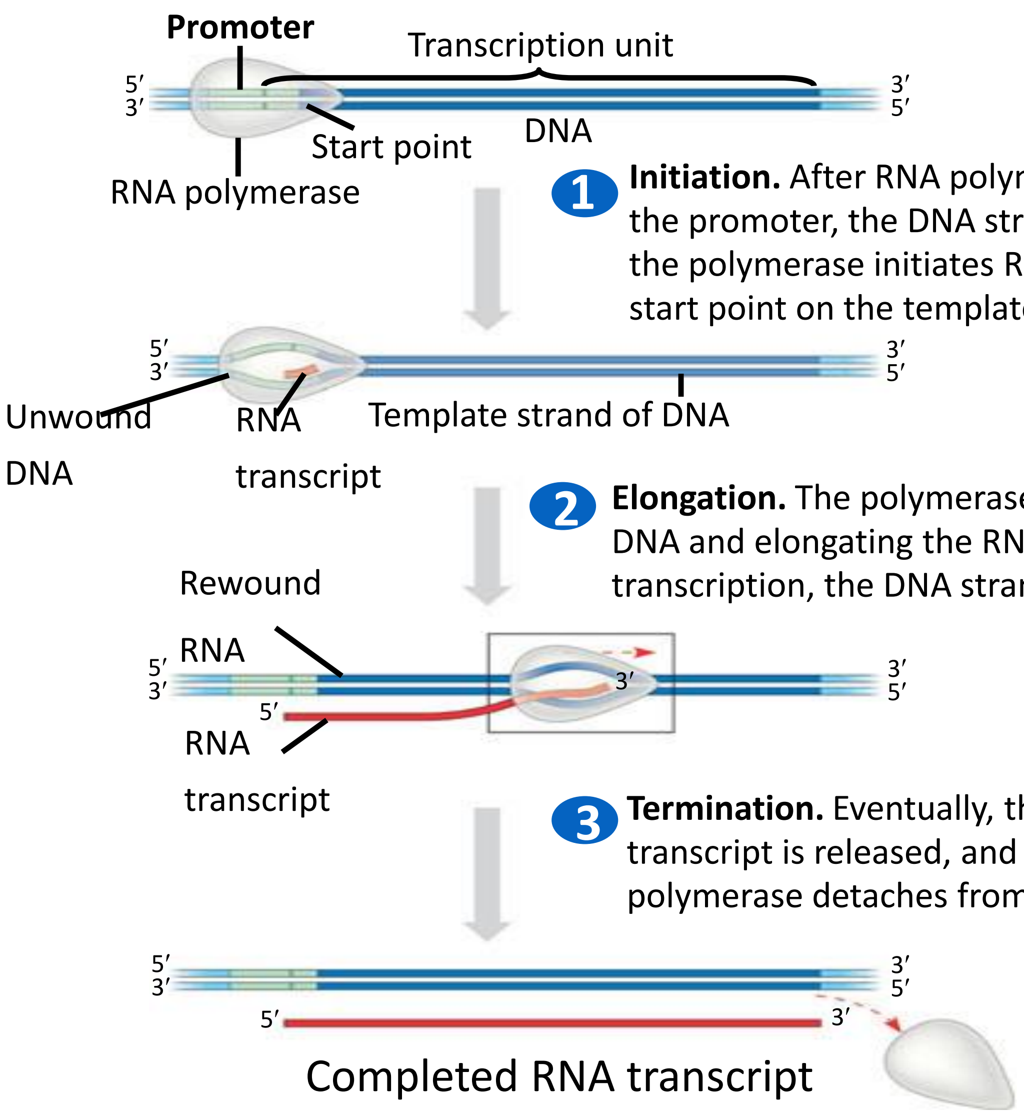
- **mRNA (messenger RNA)** carries the information encoded in DNA out of the nucleus to the protein assembly machinery, called the ribosome, in the cytoplasm.
- **mRNA transcription** - the synthesis of mRNA from a DNA
  - coding portions of a gene, exons, interrupted by introns
  - editing process removes the introns, to make a **mature mRNA**

mRNA transcription process





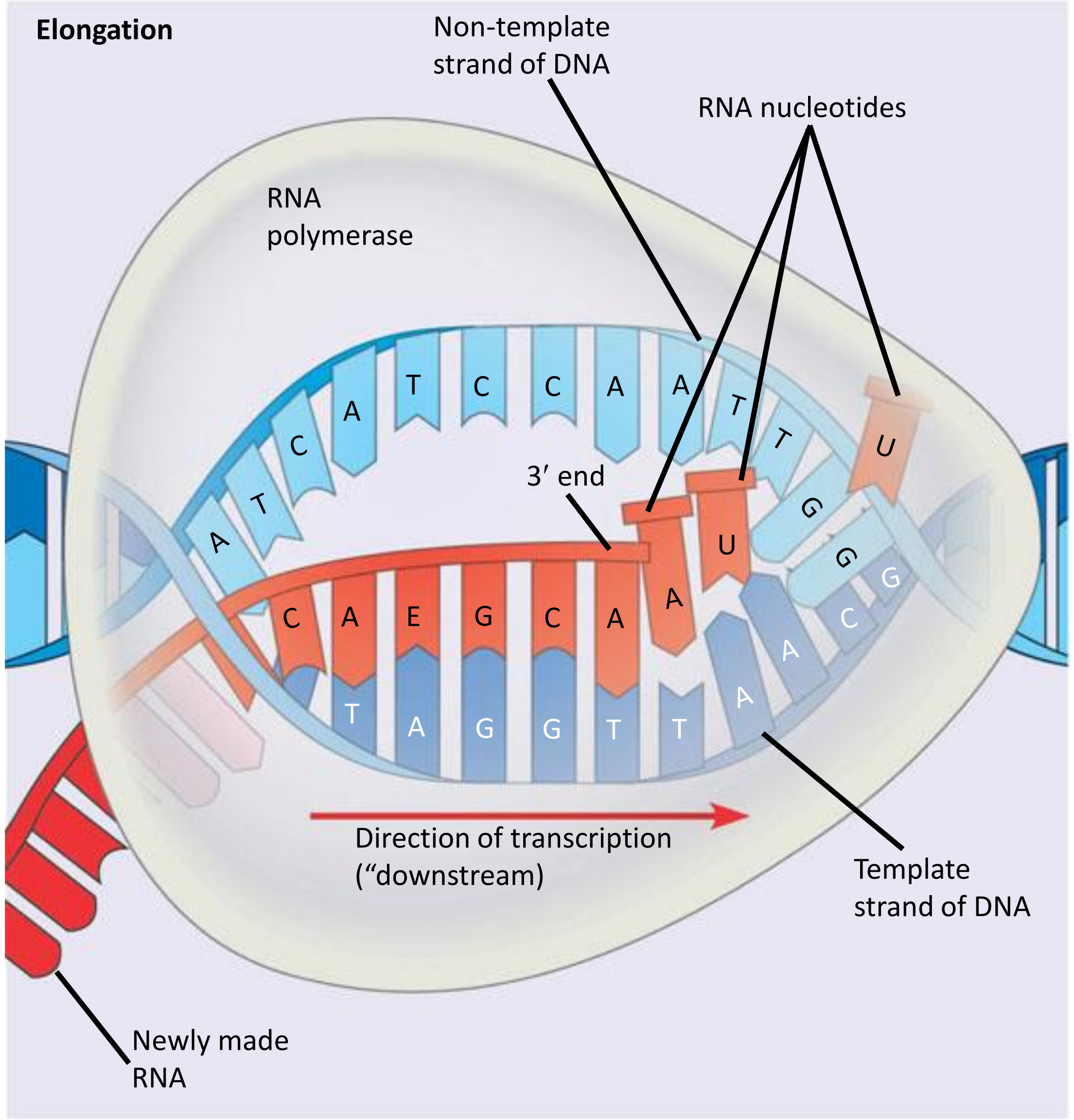
Initiation, elongation, and termination



1 **Initiation.** After RNA polymerase binds to the promoter, the DNA strands unwind, and the polymerase initiates RNA synthesis at the start point on the template strand.

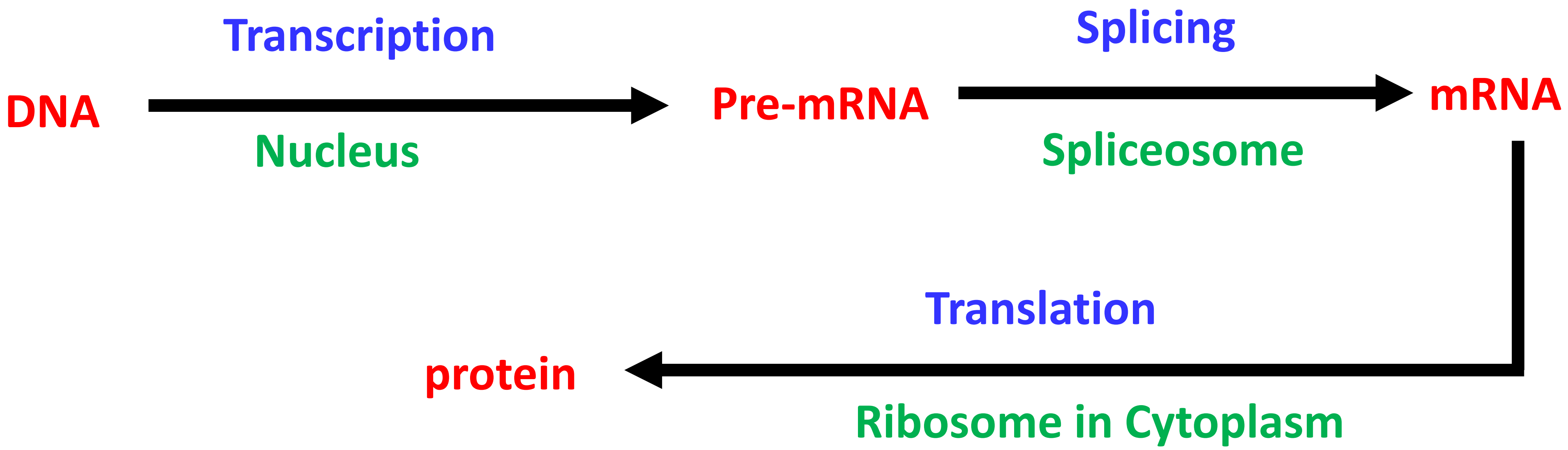
2 **Elongation.** The polymerase moves downstream, unwinding the DNA and elongating the RNA transcript 5′ → 3 ′. In the wake of transcription, the DNA strands re-form a double helix.

3 **Termination.** Eventually, the RNA transcript is released, and the polymerase detaches from the DNA.



Elongation
Non-template strand of DNA
RNA nucleotides
RNA polymerase
3′
5′
5′
3′ end
A T C A T C C A A T T G G
C A E G C A A U U
T A G G T T A A C G
Direction of transcription ("downstream)
Template strand of DNA
Newly made RNA

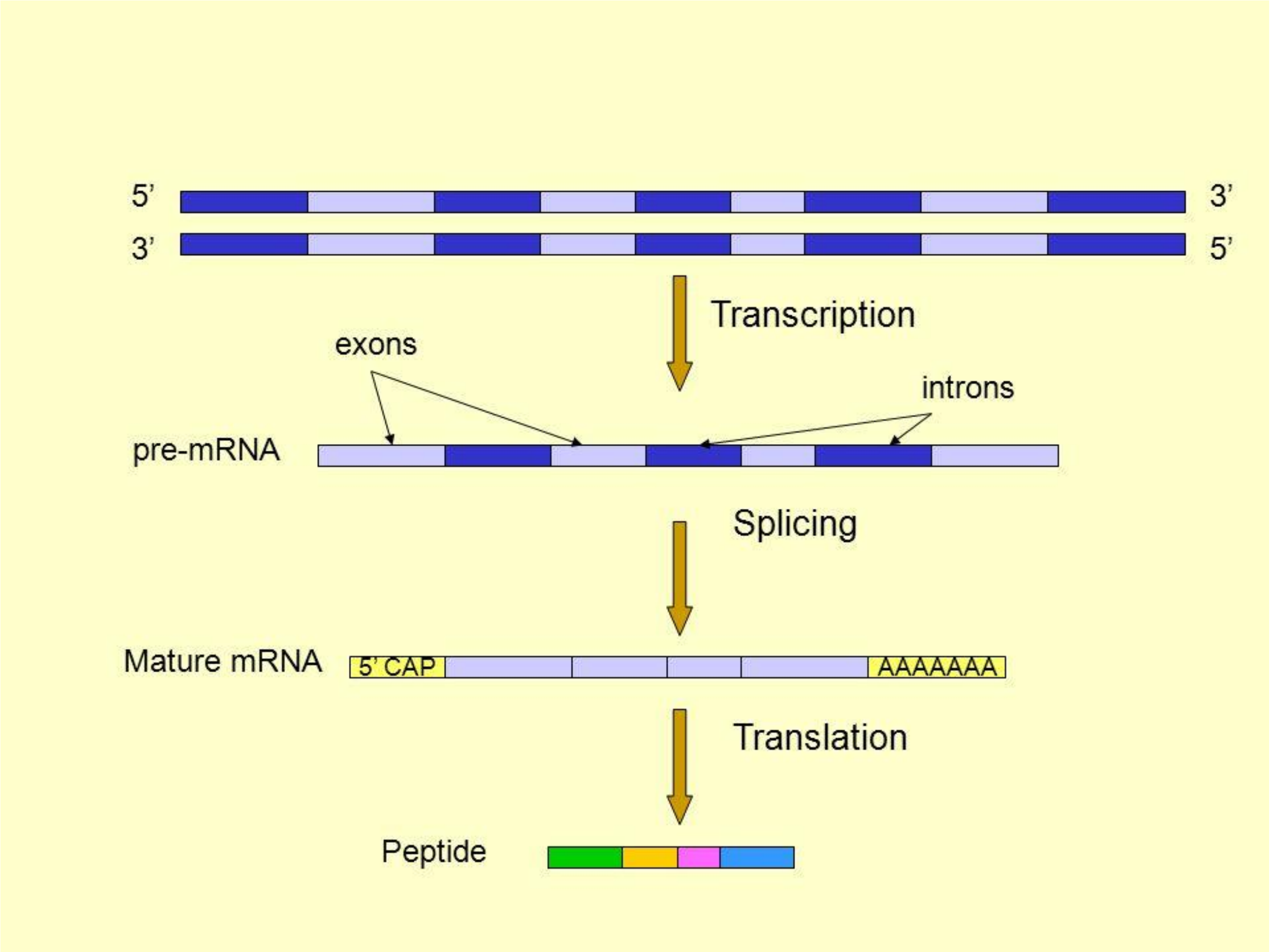
Central dogma revisited



- **Base Pairing Rule**: A and T (or U) is held together by 2 hydrogen bonds and G and C is held together by 3 hydrogen bonds
- **Note**: Some mRNA stays as RNA (e.g. tRNA, rRNA)

Splicing

- Splice variants create an enormous diversity of proteins
  - ~25,000 genes in humans give rise to 200,000 to 2,000,000 different proteins
  - Splice variants may have very diverse functions



Alternative splicing

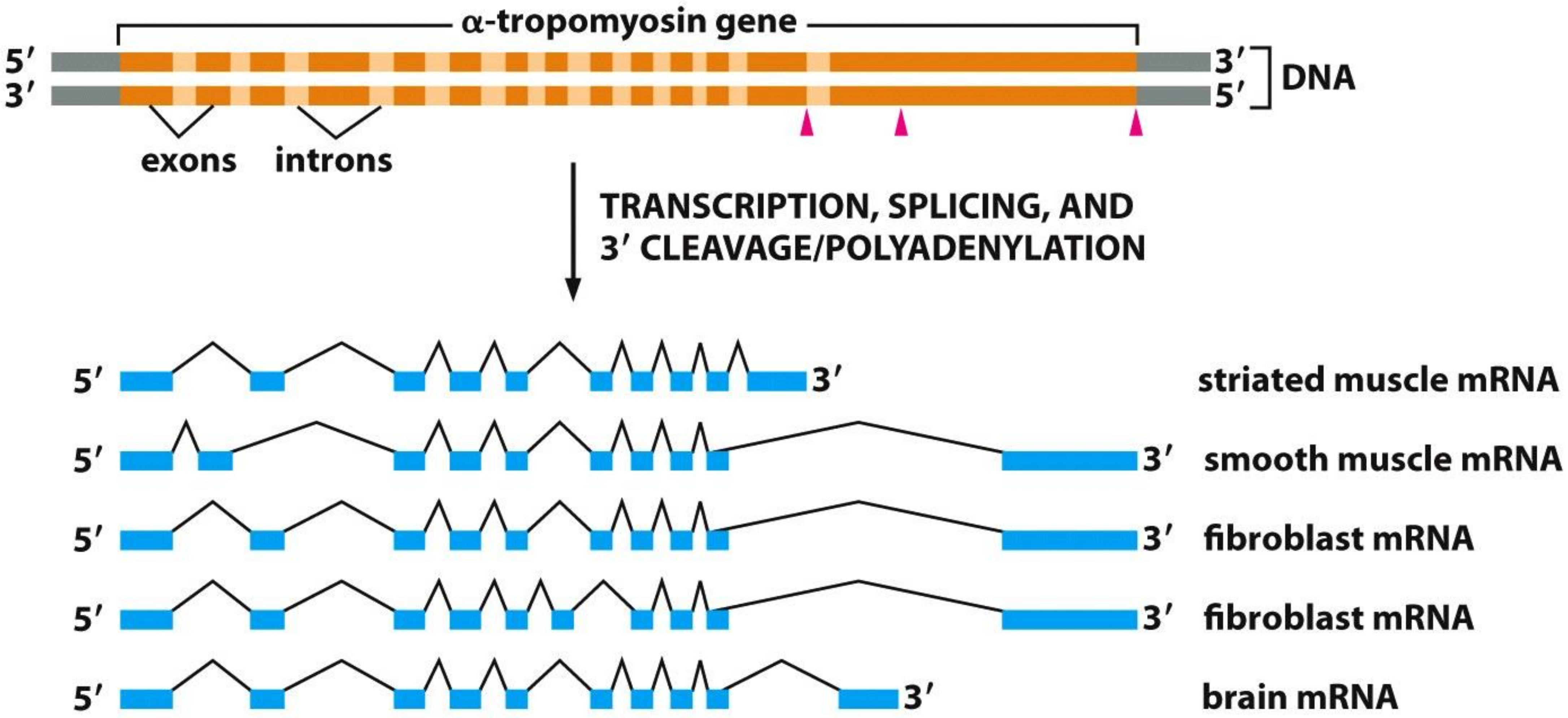


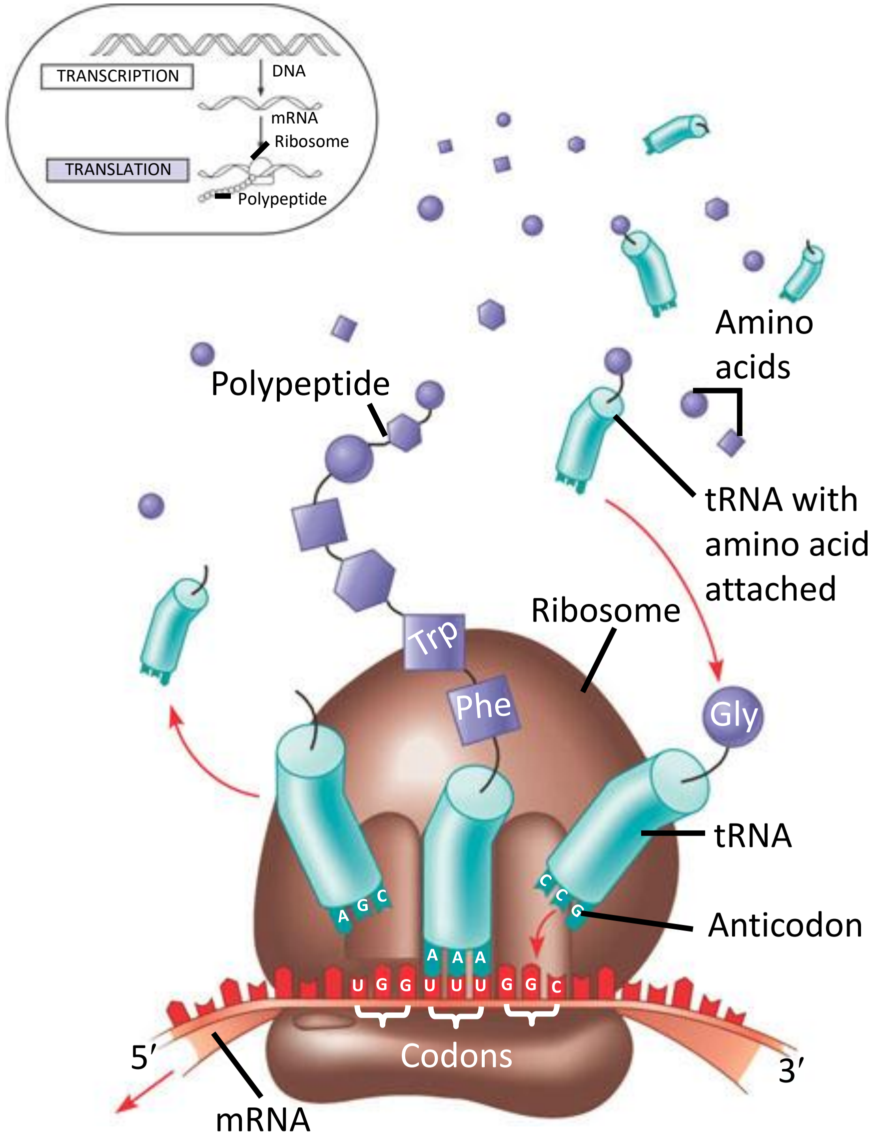
Figure 7-21 Essential Cell Biology 3/e (© Garland Science 2010)

Protein Translation

Protein translation

- DNA - carrier of genetic information
- Protein - do the bulk of the work in the cell
  - A long chain containing as many as 20 different kinds of amino acids
  - The order & number of amino acids determine the shape and function of the protein
- Each cell contains thousands of different proteins
- The ribosomal complex builds a protein one amino acid at a time, with the order of amino acids determined precisely by the order of the **codons** in the mRNA

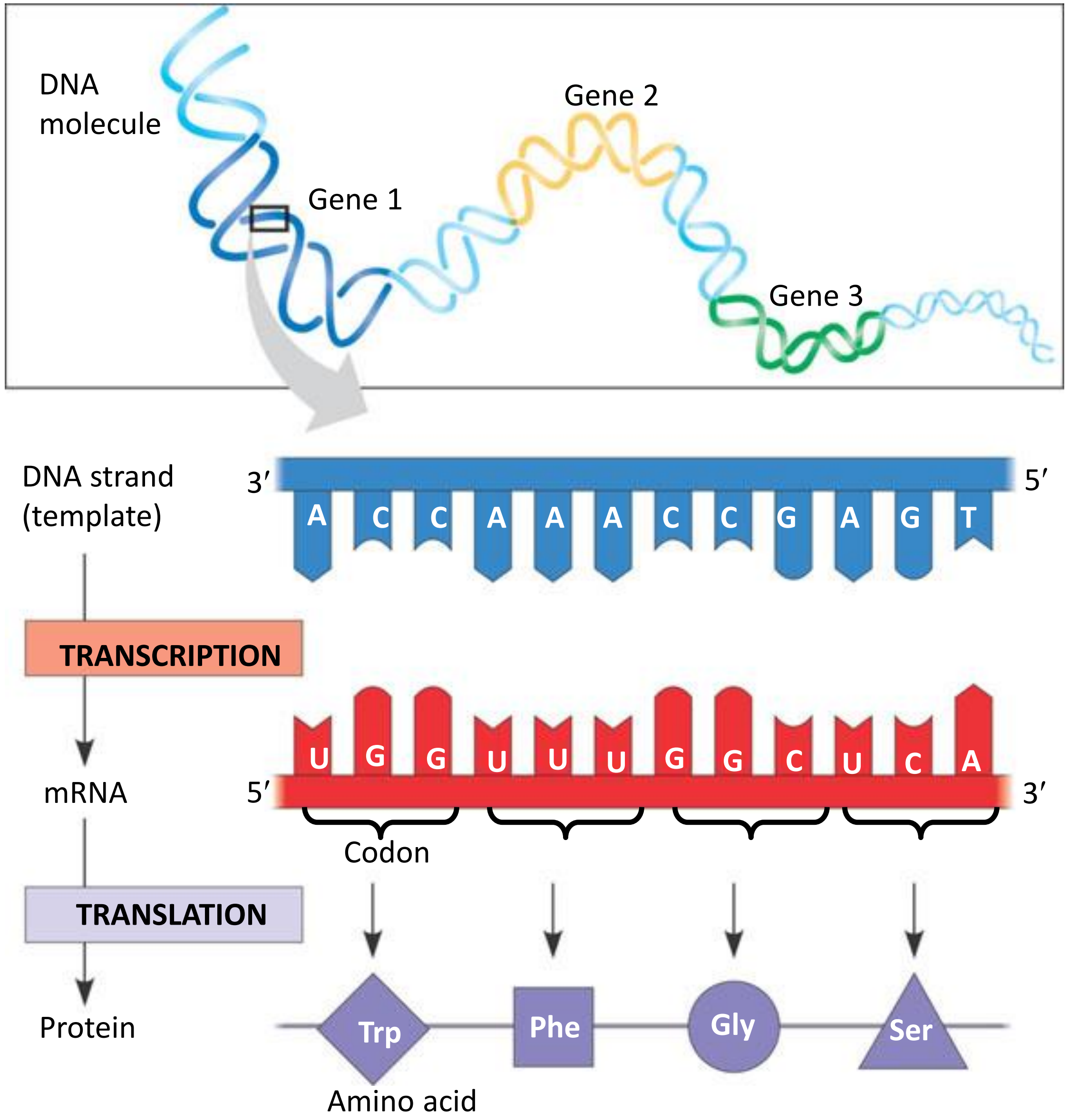
Translation: the basic concept



Terminology

- **Codon**: The sequence of 3 nucleotides in DNA/RNA that encodes for a specific amino acid
- **mRNA (messenger RNA)**: A ribonucleic acid whose sequence is complementary to that of a protein-coding gene in DNA
- **Ribosome**: The organelle that synthesizes polypeptides under the direction of mRNA
- **rRNA (ribosomal RNA)**: The RNA molecules that constitute the bulk of the ribosome, provide structural scaffolding for the ribosome and catalyze peptide bond formation
- **tRNA (transfer RNA)**: The small L-shaped RNAs that deliver specific amino acids to ribosomes according to the sequence of a bound mRNA

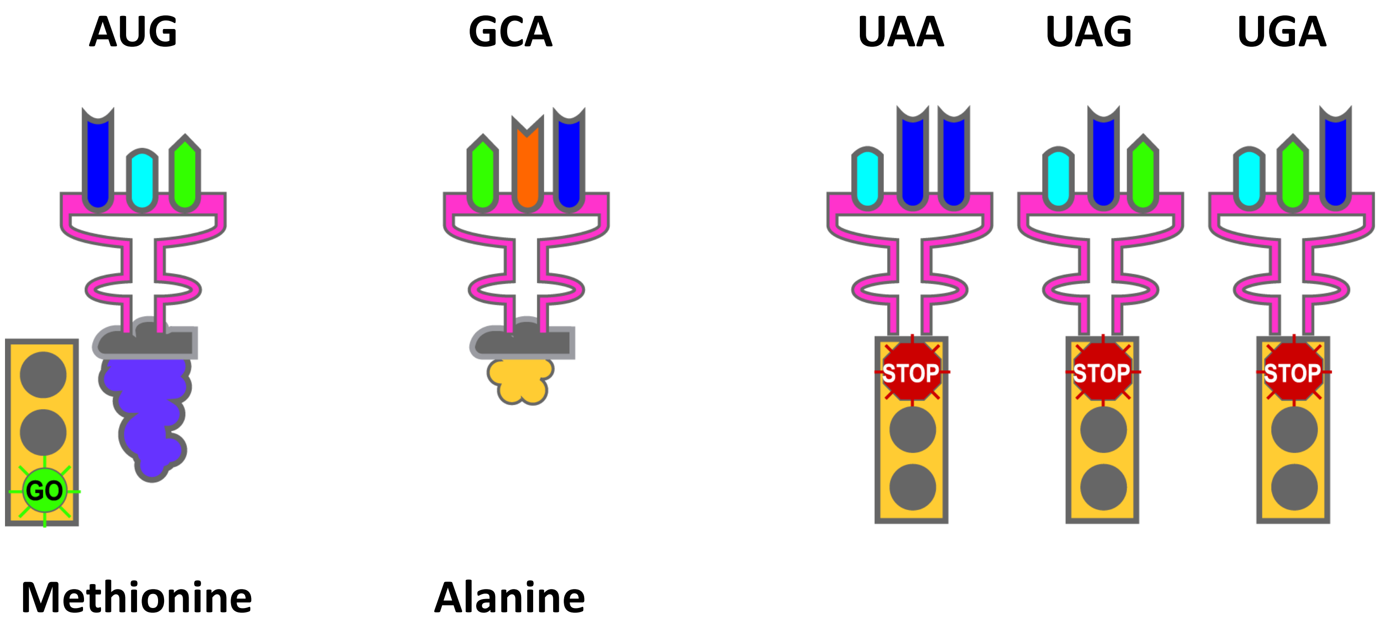
The triplet code



64($=4^3$) RNA triplet codons and their corresponding amino acids

		2nd base			
		U	**C**	**A**	**G**
1st base	**U**	UUU (Phe/F)Phenylalanine UUC (Phe/F)Phenylalanine UUA (Leu/L)Leucine UUG (Leu/L)Leucine	UCU (Ser/S)Serine UCC (Ser/S)Serine UCA (Ser/S)Serine UCG (Ser/S)Serine	UAU (Tyr/Y)Tyrosine UAC (Tyr/Y)Tyrosine UAA Ochre (*Stop*) UAG Amber (*Stop*)	UGU (Cys/C)Cysteine UGC (Cys/C)Cysteine UGA Opal (*Stop*) UGG (Trp/W)Tryptophan
	C	CUU (Leu/L)Leucine CUC (Leu/L)Leucine CUA (Leu/L)Leucine CUG (Leu/L)Leucine	CCU (Pro/P)Proline CCC (Pro/P)Proline CCA (Pro/P)Proline CCG (Pro/P)Proline	CAU (His/H)Histidine CAC (His/H)Histidine CAA (Gln/Q)Glutamine CAG (Gln/Q)Glutamine	CGU (Arg/R)Arginine CGC (Arg/R)Arginine CGA (Arg/R)Arginine CGG (Arg/R)Arginine
	A	AUU (Ile/I)Isoleucine AUC (Ile/I)Isoleucine AUA (Ile/I)Isoleucine AUG (Met/M)Methionine, *Start*[1]	ACU (Thr/T)Threonine ACC (Thr/T)Threonine ACA (Thr/T)Threonine ACG (Thr/T)Threonine	AAU (Asn/N)Asparagine AAC (Asn/N)Asparagine AAA (Lys/K)Lysine AAG (Lys/K)Lysine	AGU (Ser/S)Serine AGC (Ser/S)Serine AGA (Arg/R)Arginine AGG (Arg/R)Arginine
	G	GUU (Val/V)Valine GUC (Val/V)Valine GUA (Val/V)Valine GUG (Val/V)Valine	GCU (Ala/A)Alanine GCC (Ala/A)Alanine GCA (Ala/A)Alanine GCG (Ala/A)Alanine	GAU (Asp/D)Aspartic acid GAC (Asp/D)Aspartic acid GAA (Glu/E)Glutamic acid GAG (Glu/E)Glutamic acid	GGU (Gly/G)Glycine GGC (Gly/G)Glycine GGA (Gly/G)Glycine GGG (Gly/G)Glycine

Types of codons



Protein structure and function

- A protein is a **polypeptide**. However, it is a very difficult problem to understand the functions of a protein given only the polypeptide sequence
- **Protein folding** is an open problem. The 3D structure depends on many variables
- Improper folding of a protein is believed to be the cause of diseases, e.g. mad cow disease

Hierarchy of protein structures

```
DNA sequence      : 5′- ATG CCC TGC TTG GCC …- 3′
RNA sequence      : 5′- AUG CCC UGC UUG GCC …-3′
Protein sequence  :      M   P   C   L   A  …
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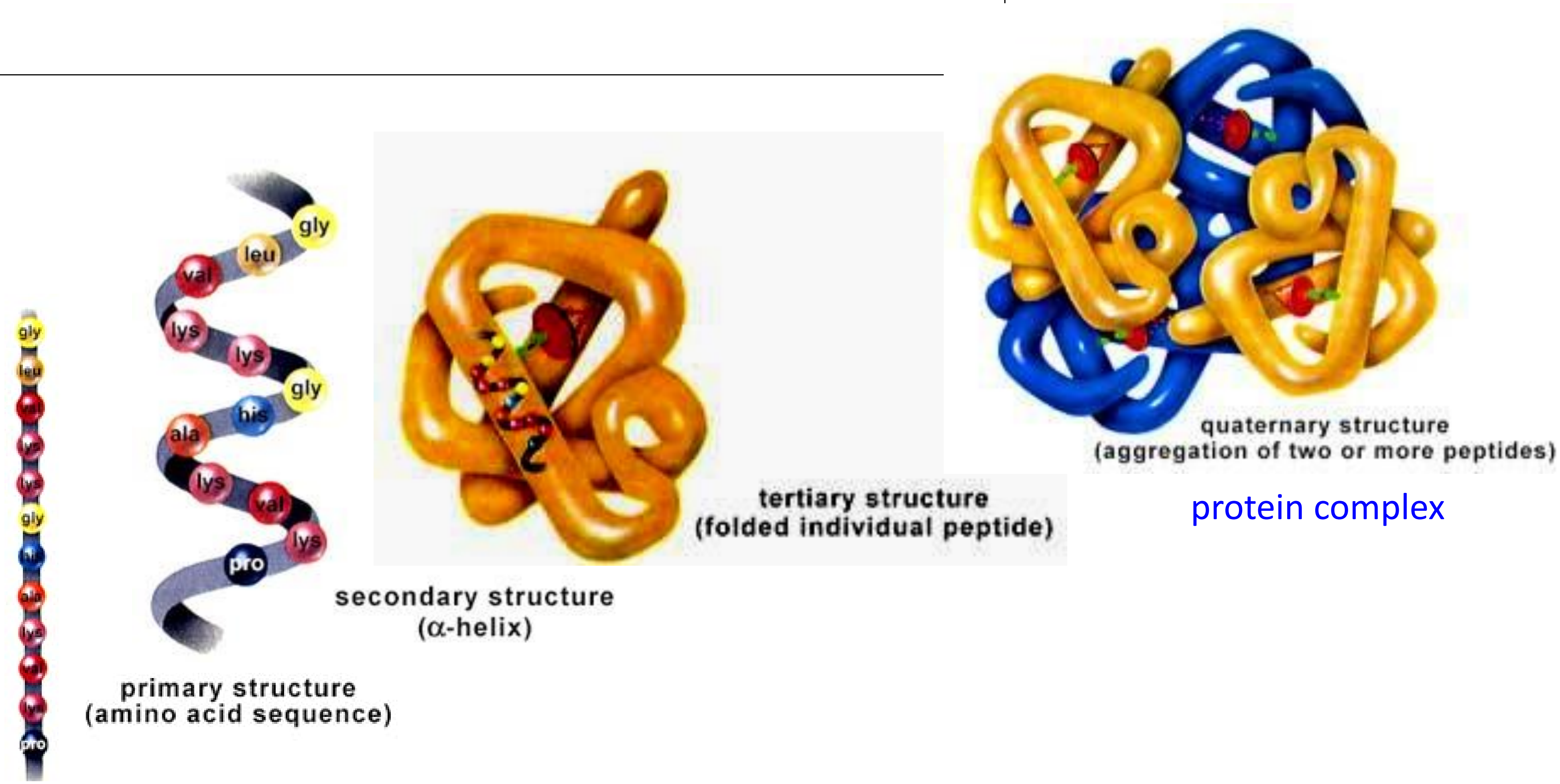
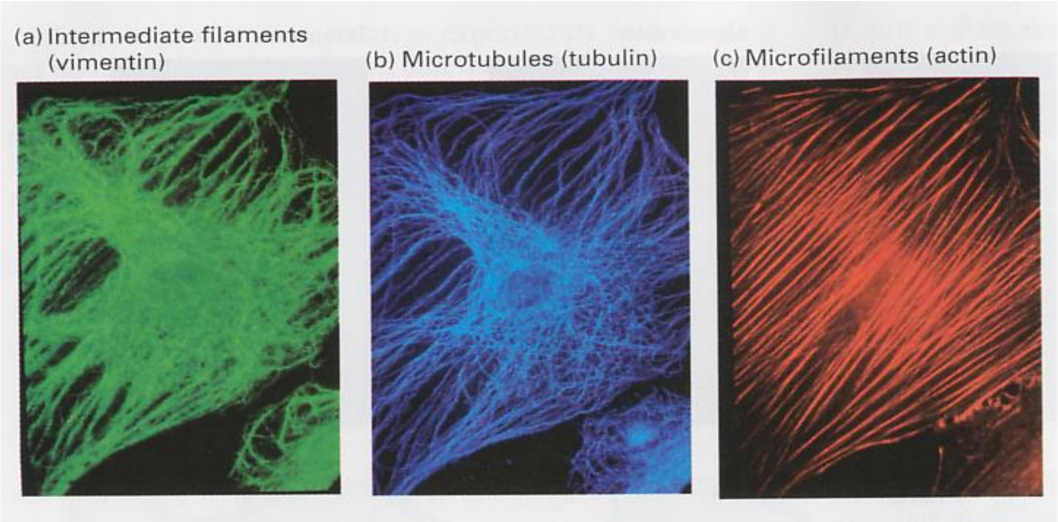
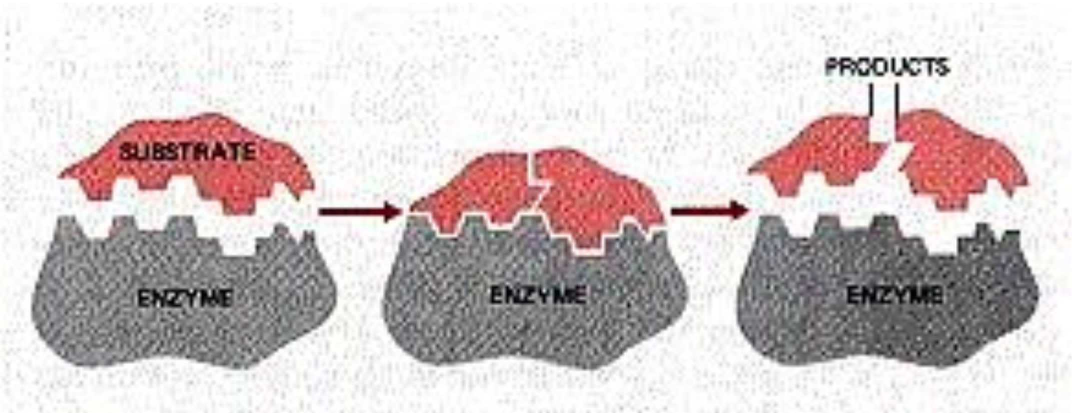


protein complex

Protein functions



e.g. Structural role of proteins



e.g. Proteins as enzymes (catalysts)

Recap

- In this lecture, we have learned:
  - The historical and intellectual roots of molecular biology
  - What is life made of?
  - The Central Dogma of Molecular Biology
  - How are RNA produced by transcription?
  - How are proteins produced by translation?

References

- E. Lander and M. Waterman (1995). "**The Secrets of Life: A Mathematician's Introduction to Molecular Biology**", Chapter 1 of the National Research Council report entitled *"Calculating the Secrets of Life: Contributions of the Mathematical Sciences to Molecular Biology"*, National Academy Press, Washington D.C., 1995.
- N. Jones and P. Pevzner (2004). "**Molecular Biology Primer**", Chapter 3 of their book *"An Introduction to Bioinformatics Algorithms"*, MIT Press, 2004.
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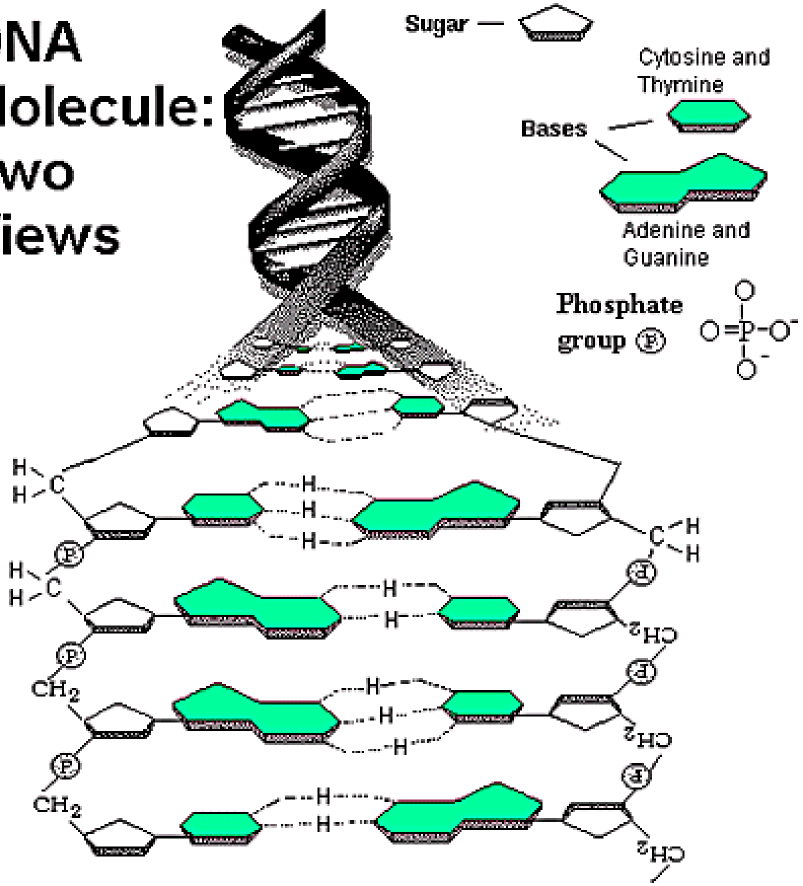
Appendix A: Replication and Reproduction

DNA replication: complementary strands

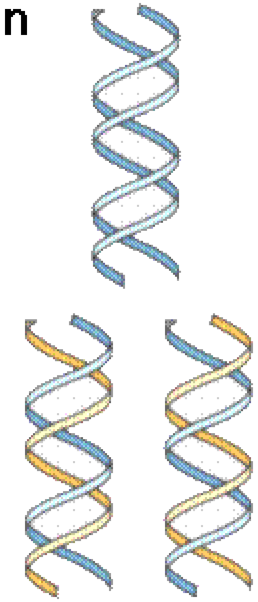
DNA Molecule: Two Views



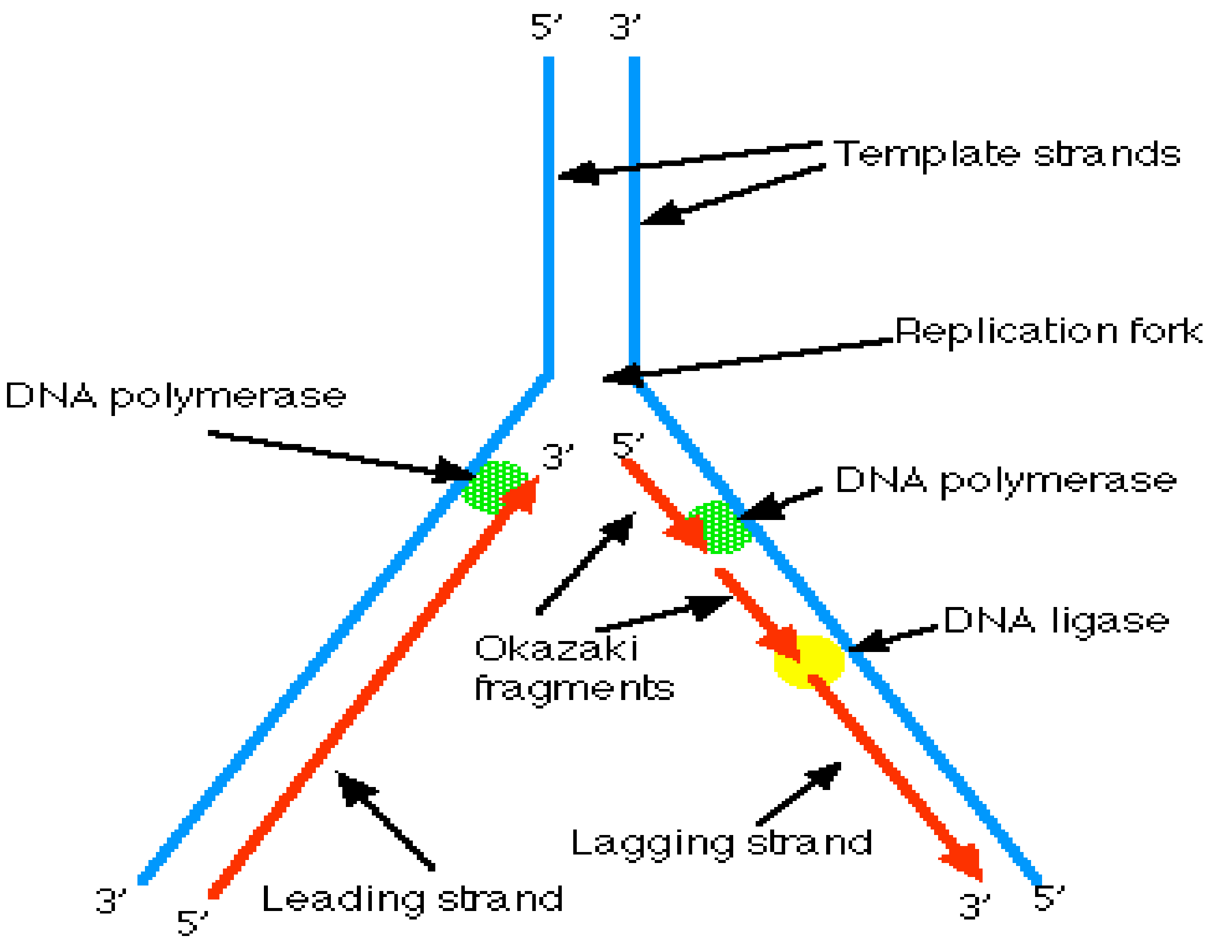
Semiconservative replication

Original DNA Helix

DNA helixes after one round of replication

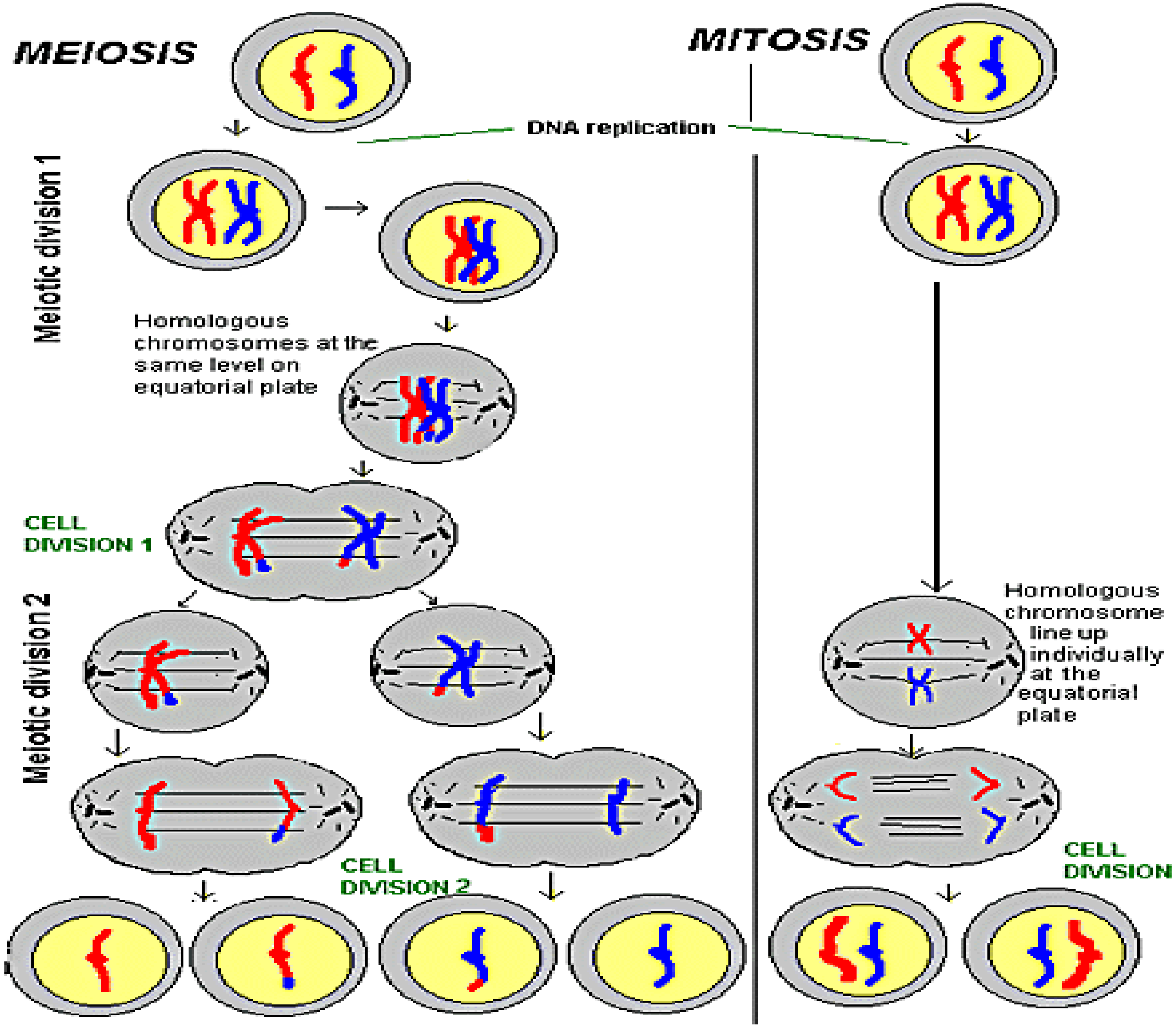


DNA replication process



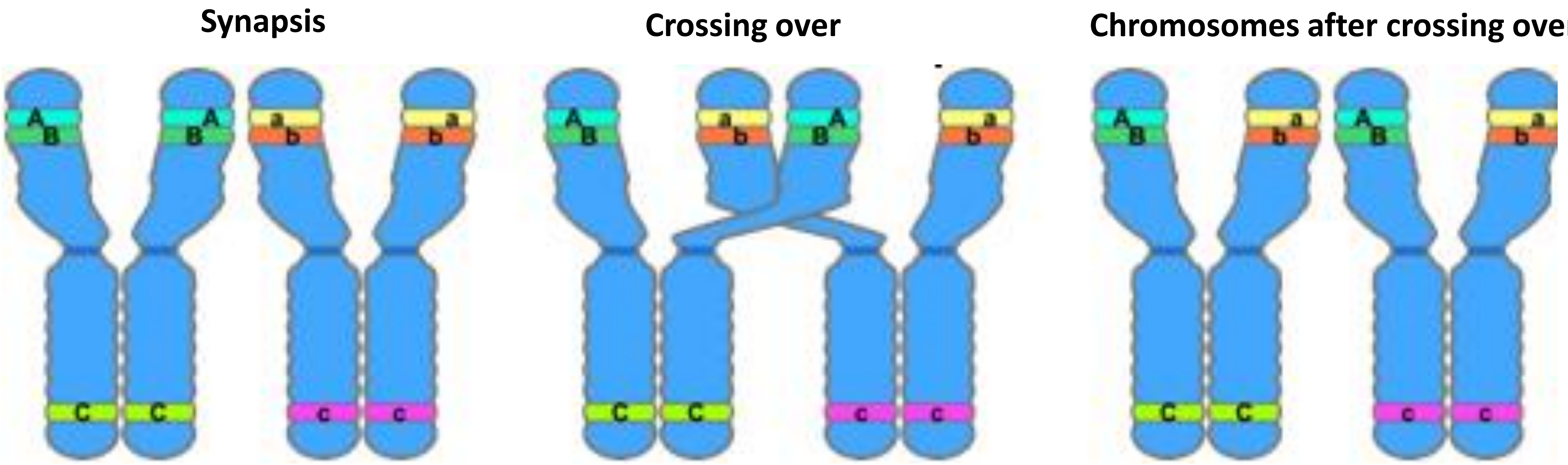
Sexual reproduction

- To reproduce, eukaryotes must first create special cells called gametes (eggs and sperms) that then fuse to form the beginning of a new organism.
- Meiosis is a specialized type of cell division that occurs during the formation of gametes. It is really just two cell divisions in sequence. Each of these sequences maintains strong similarities to mitosis.
  - Meiosis serves to reduce the chromosome number for that particular organism by half.


MEIOSIS
MITOSIS
DNA replication
Meiotic division 1
Homologous chromosomes at the same level on equatorial plate
CELL DIVISION 1
Meiotic division 2
CELL DIVISION 2
Homologous chromosome line up individually at the equatorial plate
CELL DIVISION

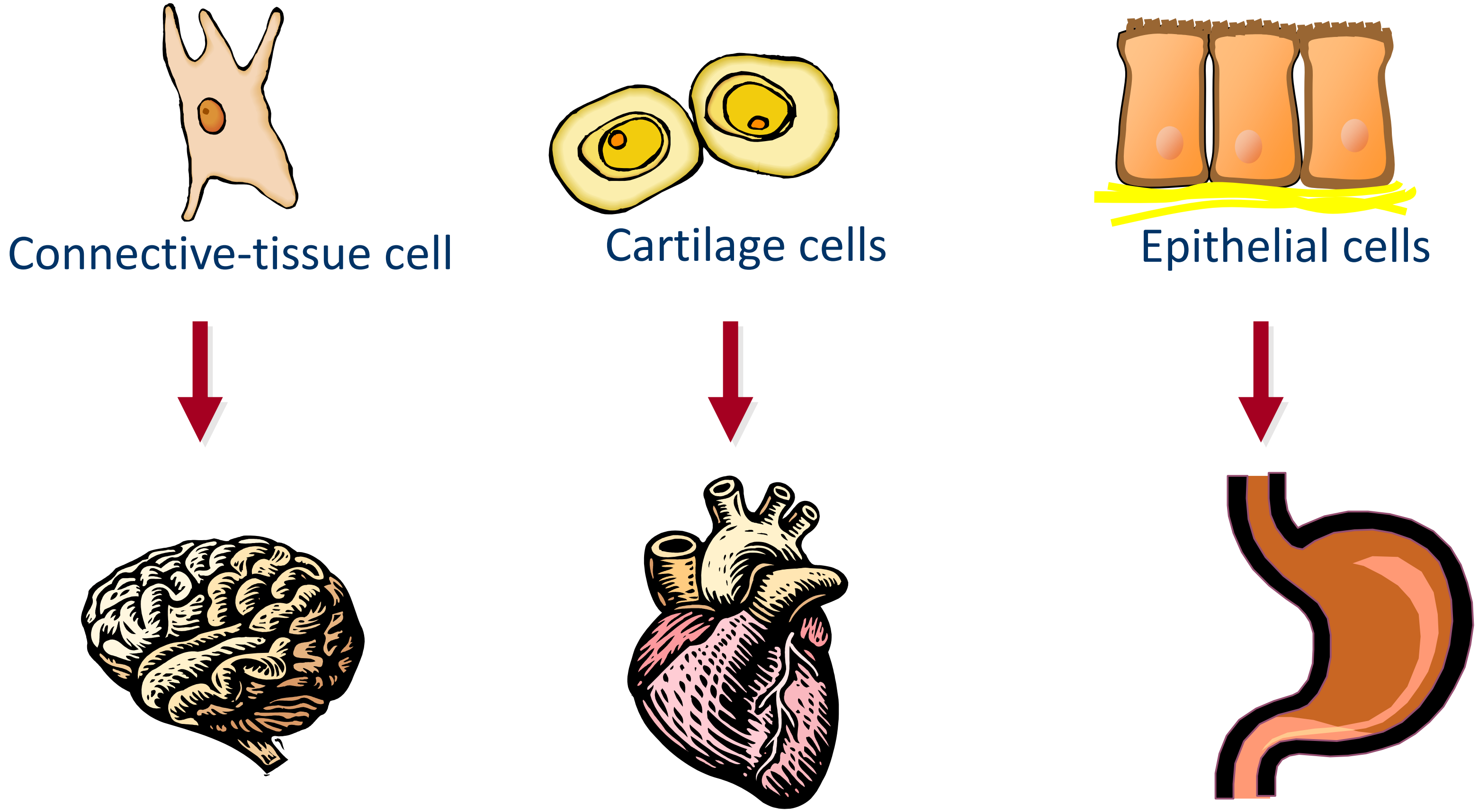
Genetic recombination - diversity

- Genetic recombination refers to a large-scale rearrangement of a DNA molecule.



Cell differentiation

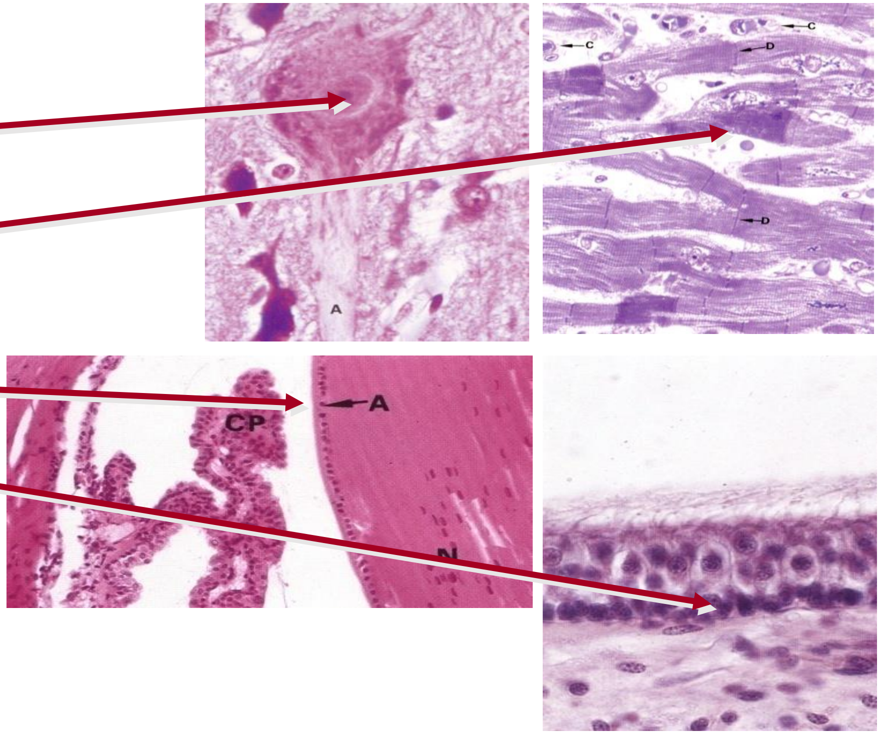
- **Differentiation** is the process by which an unspecialized cell becomes specialized into one of the many cells that make up the body



Cell differentiation (continued)

- During differentiation, certain genes are turned on, or become activated, while other genes are switched off, or inactivated
- This process is intricately regulated

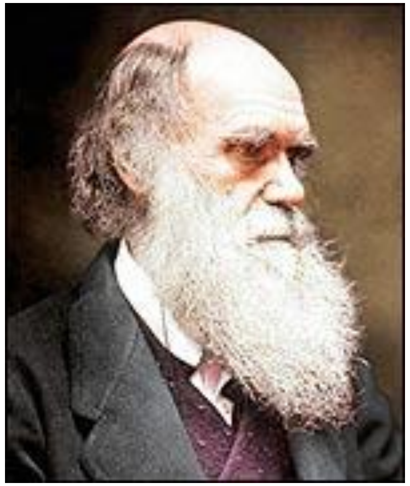
- e.g.
  Nerve cells
  Cardiac muscle cells
  Lens cells of the eye
  Auditory hair cells of the ear



Appendix B: Genetic Variation

What is the difference between man and ape?

- Man and chimpanzee have a genomic similarity of greater than 95%
- What accounts for differences between species?



Physical traits and variances

- Individual variation among a species occurs in populations of all sexually reproducing organisms
- Individual variations range from hair and eye color to less subtle traits such as susceptibility to malaria
- Physical variation is the reason we can pick out our friends in a crowd. However, most physical traits and variation can only be seen at a cellular and molecular level

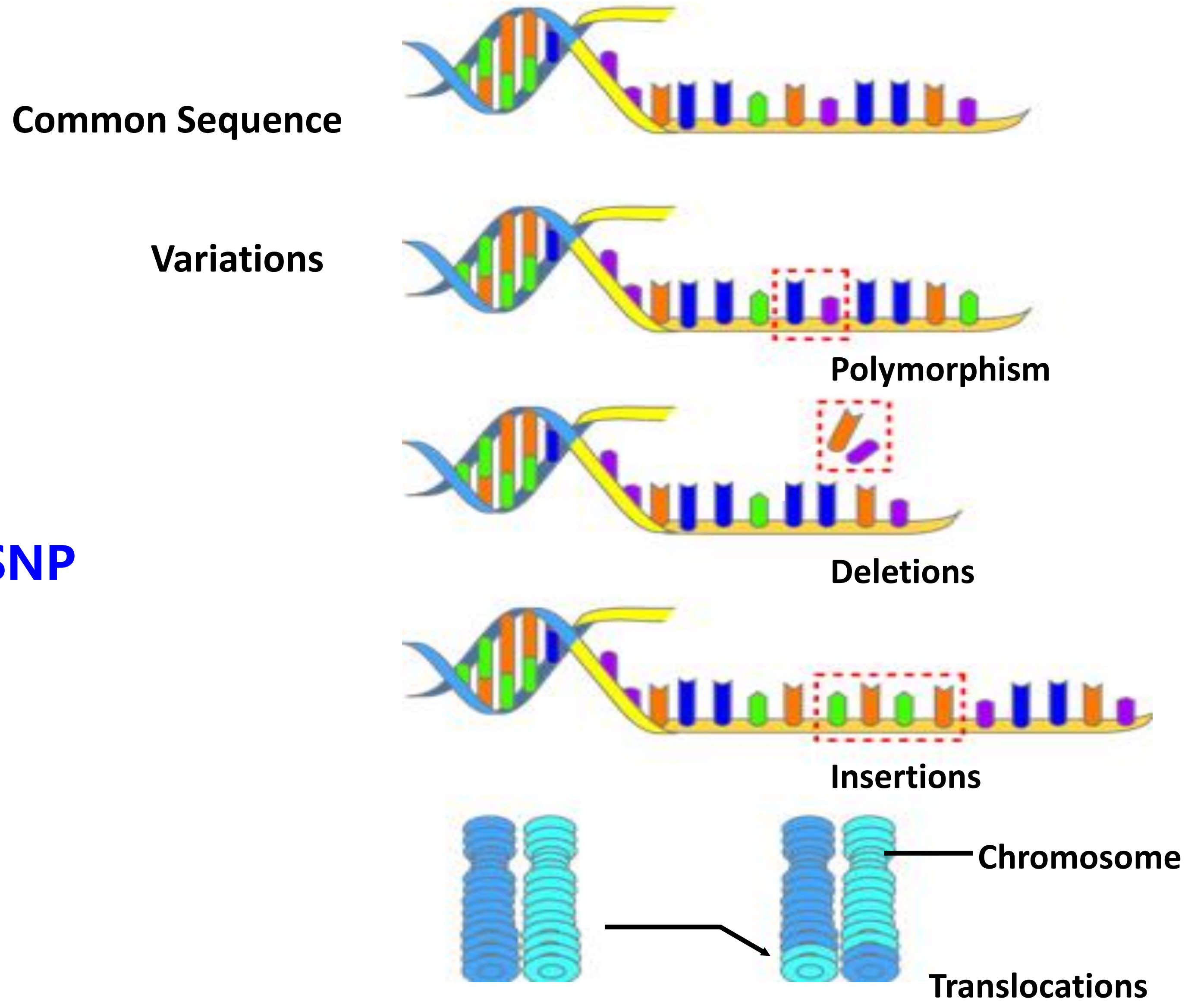
Genetic variation

- Despite the wide range of physical variation, genetic variation between individuals is quite small
- Out of 3 billion nucleotides, only roughly 3 million base pairs (0.1%) are different between individual genomes of humans
- Although there is a finite number of possible variations, the number is so high ($4^{3,000,000}$) that we can assume no two individual people have the same genome
- What is the cause of this genetic variation?

Sources of genetic variation

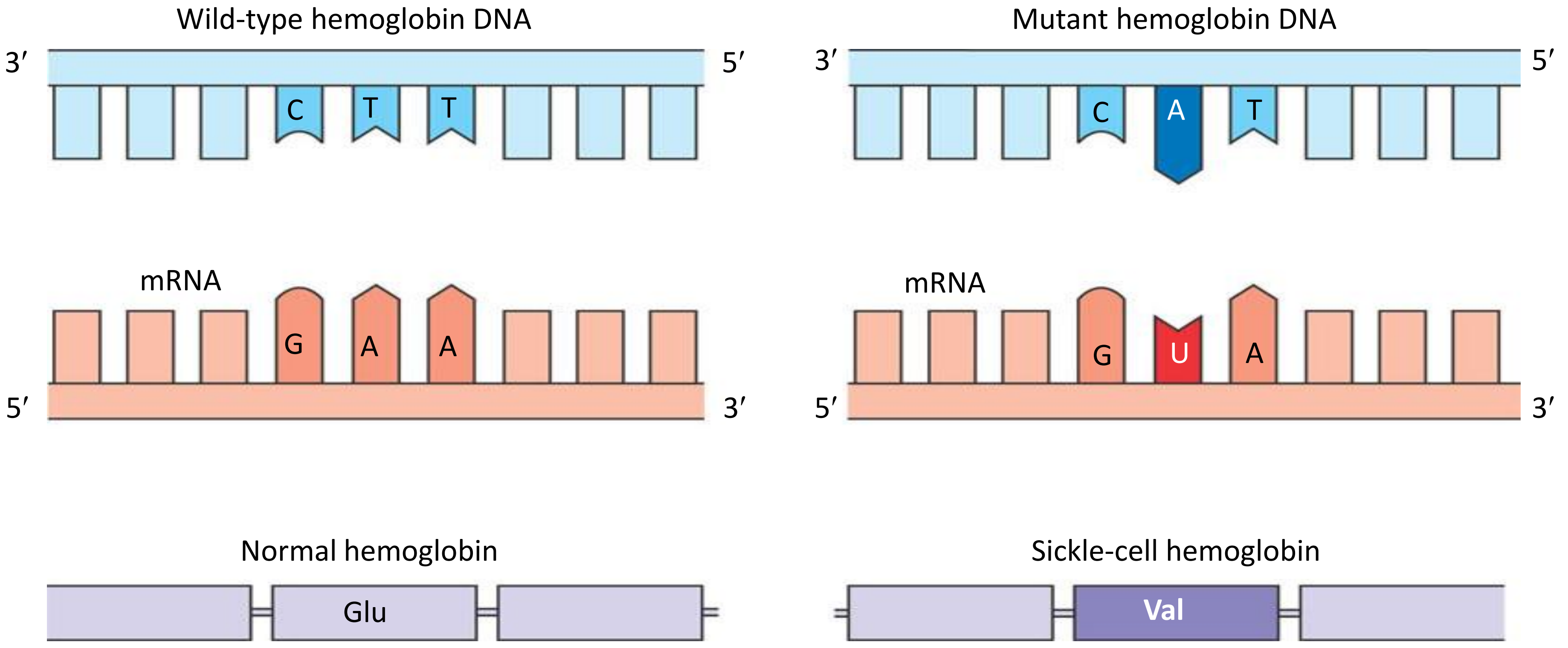
- **Mutations** are rare errors in the DNA replication process that occur at random
- When mutations occur, they affect the genetic sequence and create genetic variation between individuals
- Most mutations do not create beneficial changes and actually kill the individual
- Although mutations are the source of all new genes in a population, they are so **rare** that there must be another process at work to account for the large amount of diversity

Variation through mutation



- **Single Nucleotide Polymorphism** or **SNP**

The molecular basis of sickle-cell disease: a point mutation



In the DNA, the mutant template strand has an A where the wild-type template has a T.

The mutant mRNA has a U instead of an A in one codon.

The mutant (sickle-cell) hemoglobin has a valine (Val) instead of a glutamic acid (Glu).

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More sources of genetic variation

- **Recombination** is the shuffling of genes that occurs through sexual mating and is the main source of genetic variation
- Recombination occurs via a process called **crossing over** in which genes switch positions with other genes during meiosis
- Recombination means that new generations inherit random combinations of genes from both parents
- The recombination of genes creates a seemingly endless supply of genetic variation within a species
- Other sources: genome rearrangement, gene duplication, etc.